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SECTION 3

Executive Summary

What changed from v13.1 FINAL to v14.0 Clean Master Edition, and why.

The Clean Master Edition v14.0 is a structural reorganisation of the SGTX Platform Master Blueprint. It does not introduce new architectural commitments; it clarifies, separates, and audits the commitments already made in v13.1 FINAL and its source documents (v12.0 Main Blueprint and the Add-Ons Change-Set covering Add-Ons 1 through 28).

Purpose of this Edition

v13.1 FINAL was an integrated, audited consolidation of the SGTX platform specification. It carried the full v12.0 Main Blueprint, the v12.0 audit, the v13.0 re-audit (A-01 through A-24), the full twenty-eight-add-on catalogue, and the SGTX Master Global Trade Completion Amendment. While complete, it suffered from three structural weaknesses: (i) constitutional principles, normative architecture, and implementation specifications were intermixed throughout the body, making it difficult for an auditor to determine which statements were immutable; (ii) the document retained legacy over-claim vocabulary (most notably *production-ready*) that the v13.0 audit had already resolved semantically but not yet eliminated textually; and (iii) the deployment-state vocabulary established by finding A-19 was applied inconsistently. v14.0 resolves all three weaknesses.

The Three-Layer Separation

The defining change in v14.0 is the introduction of an explicit three-layer separation. Every normative statement in the document is now tagged with its layer:

- **[L0] Layer 0 — Constitution.** Immutable principles that can only be amended through the formal constitutional amendment process (multisig of constitutional signatories plus a notice period). These principles include the seven Governor gates (G1 through G7), the thirty-two-point SGTX Transaction Constitution, the AI Authority Ladder (A0 through A5), the Non-Custody Principle, the Non-Marketplace Principle, and the amendment process itself.
- **[L1] Layer 1 — Architecture.** The current normative architecture of the SGTX platform. These are the structural commitments that hold today: the multi-clock state vector model, the event spine, the Governor pipeline, the settlement orchestration control plane, the twenty-eight-add-on catalogue with its status matrix, the jurisdiction capability adapter schema, the regulatory classification gate, the closure policy, the AI recommendation gateway, and the transport engine architecture. Architectural commitments may evolve without constitutional amendment, but their evolution is itself governed by L0.
- **[L2] Layer 2 — Implementation Specifications.** The canonical event type catalogue, the canonical data model outline, the API contract structure, the observability catalogue, and the RTO/RPO targets. These are concrete engineering specifications — they are expected to be implemented and verified against, but their textual form is intended to be refined as implementation proceeds.

Section 3 | Executive Summary

Over-Claim Elimination

Finding A-19 of the v13.0 audit established a deployment-state vocabulary to replace the broad *production-ready* language carried over from the v12.0 Main Blueprint. v14.0 completes this resolution textually. The forbidden phrases and their canonical replacements are:

Forbidden Phrase	Canonical Replacement	Rationale
production-ready	CORE_READY or PRODUCTION_CONNECTED	Distinguishes internal architectural readiness from external operational readiness
complete	specified	Avoids implying implementation completeness where only specification exists
zero-cost	institutional-cost scope clarified	Acknowledges that all integrations carry institutional cost
fully integrated	CORE_READY, PRODUCTION_CONNECTED, legally authorised	A capability is production-operational only when all three conditions hold
globally available	jurisdiction-aware, capability-gated	Availability is per-jurisdiction and per-capability, not global

Deployment-State Vocabulary

The deployment-state vocabulary established by finding A-19 and formalised in v14.0 consists of three primary states and a four-dimension external readiness model. A capability is production-operational only when all of the following hold:

State	Definition	Evidence Required
CORE_READY	The internal architecture for the capability is specified, implemented, and verified against the L2 implementation specifications.	Passing test suite; L2 conformance report; Governor integration test.
PRODUCTION_CONNECTED	The capability is wired to a real, live external system (government API, bank rail, customs gateway, carrier system) and round-trip transactions have been observed in production-equivalent environments.	Successful end-to-end transactions; signed connector attestation; operational runbook.
LEGAL_AUTHORIZATION_REQUIRED	The capability requires explicit legal authorisation (licence, registration, regulatory clearance) per the relevant jurisdiction before it can be activated for production traffic.	Jurisdiction-specific licence; regulatory filing receipt; legal opinion.

The four-dimension external readiness model (TECHNICAL, LEGAL, OPERATIONAL, COMMERCIAL) is described in detail in Layer 1, Part 6 (Jurisdiction Capability Adapter Schema) [see Part 6]. The four dimensions are evaluated independently per jurisdiction; no dimension may be inferred from another.

Audit Trail Preservation

A central commitment of v14.0 is that the full v13.1 FINAL audit trail is preserved verbatim. Nothing is removed; nothing is rewritten retrospectively. The v13.1 audit findings (A-01 through A-24), the v12.0 audit header, and the complete v12.0 Main Blueprint body are all preserved in Appendix E (Full Historical Audit Trail) [see Appendix E]. The change log from v13.1 to v14.0 is itself preserved as a governing instrument in Section 8 of this document.

Command/Event Taxonomy

Finding A-20 of the v13.0 audit established the formal separation of commands from events. v14.0 elevates this to a first-class architectural principle. A command is an instruction that requests a state change; an event is an immutable record that a state change occurred (or did not occur, with reasons). Commands are mutable, may be rejected, and may be replayed; events are immutable, hash-chained, and authoritative. The full canonical event type catalogue is specified in Layer 2, Part 11 [see Part 11].

Reading Order

A reader new to the document should read Layer 0 (Constitution) first to understand the immutable principles, then Layer 1 (Architecture) to understand the current normative state of the platform, then Layer 2 (Implementation Specifications) for the concrete engineering contract. Appendices A through F provide supporting detail: the financial control framework, regulatory and legal materials, the operating model, the implementation priority framework, the full historical audit trail, and the source manifest. Section 8 (Change Log) and Section 9 (Dependency Graph) close the document.

What This Edition Does Not Do

v14.0 is not a substantive expansion of the SGTx specification. No new add-ons are introduced; no new constitutional principles are added; no existing commitments are removed. Readers seeking new architectural content should consult the v13.1 FINAL blueprint body, which is preserved in Appendix E. v14.0 is a structural and textual cleanup: the same content, reorganised for institutional defensibility.

LAYER 0

Constitution

Immutable principles. Amendment requires multisig of constitutional signatories plus a notice period. No layer below L0 may override or weaken any L0 principle.

Layer 0 Authority

Every statement in this layer is tagged [L0]. These statements are the supreme normative authority within the SGTX platform. Where any L1 architectural statement or L2 implementation statement appears to conflict with an L0 statement, the L0 statement governs and the lower-layer statement is to be read as subordinate or, where necessary, reformed. The Governor component (Layer 1, Part 3) is the operational enforcement mechanism for L0.

The Seven Governor Principles (G1 through G7)

The Governor is the constitutional enforcement component of the SGTX platform. Every state-changing operation must pass through the seven gates below before its effect is committed. The gates are sequential in normative force (an earlier-gate DENY preempts a later-gate ALLOW) and exhaustive in coverage (no operation may bypass any gate). The gates are defined as principles here at L0; their architectural realisation is specified at L1 [see Part 3].

Gat e	Principle	Operational Meaning
G1	Execution Always Gated	[L0] No state-changing operation may execute without passing through the Governor pipeline. There is no governor-bypass path, including no administrative override path.
G2	OPA Enforced	[L0] All policy decisions are expressed in OPA (Open Policy Agent) Rego modules. Policy is data; data is versioned; versions are themselves audited. No ad-hoc policy in code.
G3	WasmEdge Constitutional	[L0] The constitutional constraints (this layer) are encoded as WasmEdge modules that execute as a hard boundary around the OPA policy engine. WasmEdge may veto any OPA decision that violates the constitution.
G4	Loom Audited	[L0] Every decision (ALLOW, DENY, CONDITIONAL) and every state change is recorded in the Loom audit log. Loom entries are append-only, hash-chained, and replicated to a quorum of audit nodes. No operation may execute without a corresponding Loom entry.
G5	Multisig for Irreversible	[L0] Any operation classified as irreversible (e.g., settlement release, USTN sealing, constitutional amendment, evidence destruction) requires multisig authorisation as specified by the multisig policy [see Appendix C].
G6	AI Advisory Only	[L0] The AI subsystem may propose, classify, explain, and escalate. It may not independently authorise any state change. AI authority is bounded by the AI Authority Ladder (A0 through A5) [see A0 through A5].
G7	Bank-Authoritative Settlement	[L0] Settlement authority for fiat currency rests with the licensed bank on each payment leg. SGTX orchestrates settlement instructions; it does not execute fiat movements and does not hold funds [see Non-Custody Principle below].

Layer 0 | Constitution

The decision merge semantics across the gates are specified at L1, Part 3 [see Part 3]. In summary: a DENY at any gate overrides all ALLOW decisions from other gates; a CONDITIONAL narrows the operational space within which an ALLOW may execute; only an unopposed ALLOW across all seven gates results in execution.

The 32-Point SGTX Transaction Constitution

The thirty-two points below are the immutable constitutional constraints on every SGTX transaction. They are not policy; they are not configuration; they are not preferences. They are the constitutional floor beneath the platform. Each point is tagged [L0] and numbered C-01 through C-32 for reference. A transaction that violates any point is constitutionally invalid and must be rejected by the Governor.

#	Principle	Statement
C-01	Non-Custodial	[L0] SGTX does not take custody of funds, goods, documents, or title at any point in the transaction lifecycle. Funds remain with the licensed bank on each payment leg; goods remain with the carrier and consignee; documents remain with their authoritative issuers.
C-02	Non-Marketplace	[L0] SGTX does not operate a public matching engine. Trade relationships are relationship-controlled: a trade exists between two named, authenticated counterparties who have an established commercial relationship. SGTX does not introduce counterparties.
C-03	Non-Title-Taking	[L0] SGTX does not take title to goods at any point. Title passes between the named counterparties per the contract terms; SGTX records the title transfers as events but is never itself a title holder.
C-04	Non-Carrier	[L0] SGTX is not a carrier. Physical movement of goods is performed by licensed carriers (road, air, ocean, rail). SGTX orchestrates transport instructions and records transport events but does not itself move goods.
C-05	Non-Customs-Authority	[L0] SGTX is not a customs authority. Customs declarations are filed by licensed customs brokers; customs clearance is granted by the relevant government authority. SGTX records customs events but does not grant clearance.
C-06	Non-Bank	[L0] SGTX is not a bank and is not licensed as a bank. Fiat settlement is executed by licensed banks on each payment leg. SGTX orchestrates settlement instructions but does not itself execute fiat movements.
C-07	Non-Deposit-Taking	[L0] SGTX does not accept deposits. Funds held in connection with an SGTX transaction remain with the licensed bank; SGTX does not maintain customer accounts that hold customer funds.
C-08	Non-Government	[L0] SGTX is not a government entity. Government functions (customs, taxation, licensing, regulation) are performed by their respective authorities. SGTX interfaces with government systems but does not perform government functions.
C-09	AI-Assisted	[L0] SGTX uses AI subsystems to propose, classify, explain, and escalate. AI never independently authorises a state change. AI authority is bounded by the AI Authority Ladder [see A0 through A5].
C-10	Governor-Governed	[L0] Every state-changing operation passes through the Governor pipeline (G1 through G7). No bypass; no override; no executive override.
C-11	OPA-Enforced	[L0] All policy is expressed as OPA Rego modules. Policy is data; data is versioned; versions are themselves audited. No ad-hoc policy in code.
C-12	WasmEdge-Enforced	[L0] Constitutional constraints are encoded as WasmEdge modules. WasmEdge vetoes any OPA decision that violates this constitution. WasmEdge is the hard constitutional boundary.

Layer 0 | Constitution

#	Principle	Statement
C-1 3	Loom-Audited	[L0] Every decision and every state change is recorded in the Loom audit log. Loom is append-only, hash-chained, and replicated to a quorum of audit nodes. No operation executes without a Loom entry.
C-1 4	USTN-Centric	[L0] The Universal Sovereign Trade Number (USTN) is the canonical identifier for every trade, sub-trade, and shipment within SGTX. The USTN is a namespace, not an override of external authority [see C-15].
C-1 5	Jurisdiction-Aware	[L0] SGTX respects the jurisdictional authority of every jurisdiction in which a transaction touches. A USTN does not override customs authority, banking authority, or any other external authority. The USTN is a namespace within SGTX, not a supra-jurisdictional instrument.
C-1 6	Relationship-Control led	[L0] Trade relationships are established between named, authenticated counterparties. SGTX does not match anonymous parties. Each counterparty relationship is itself recorded as a governed artefact.
C-1 7	110% Reserve Rule	[L0] For every USTN-backed reserve instrument, the reserve ratio shall be no less than 110%. The reserve composition, attestation, and zero-knowledge evidence are specified in Appendix A [see Appendix A].
C-1 8	Closure-Is-Earned	[L0] A trade is CLOSED only when all seven closure conditions are met [see Part 8]. Closure is not granted by time elapsed; closure is earned by completion of the conditions.
C-1 9	Recovery Is Not Erasure	[L0] Recovery from a failed or partial state is permitted, but recovery does not erase the audit trail of the failure. The failed path remains in the event spine; the recovery path is a new sequence of events that builds on the failed path, not a replacement for it.
C-2 0	USTN as Namespace, Not Override	[L0] The USTN is a universal namespace within SGTX. It does not override external identifiers (carrier booking numbers, customs entry numbers, bank transaction references). External identifiers are preserved in the External Identifier Registry [see Part 12].
C-2 1	Bank-Authoritative Settlement	[L0] Fiat settlement authority rests with the licensed bank on each payment leg. SGTX orchestrates settlement instructions; the bank executes fiat movements. SGTX never holds settlement funds.
C-2 2	Command/Event Separation	[L0] Commands (requests for state change) and events (records of state change) are formally separated [see Part 2]. Commands are mutable; events are immutable. No command is also an event; no event is also a command.
C-2 3	Observation Period	[L0] After closure, a trade enters a post-closure observation period during which certain event types may still be appended to the event spine (e.g., dispute reopen, evidence supplementation) [see Section 22]. The observation period is governed; it does not permit arbitrary modification.
C-2 4	Evidence Sealing	[L0] Evidence packages for a trade are sealed at closure. Sealed evidence is content-addressable and immutable. Subsequent evidence supplementation during the observation period is recorded as a new evidence package with explicit lineage to the sealed package.
C-2 5	Multisig for Irreversible	[L0] Any operation classified as irreversible requires multisig authorisation per the multisig policy [see Appendix C]. Irreversible operations include settlement release, USTN sealing, constitutional amendment, and evidence destruction (where permitted).
C-2 6	AI Advisory Only	[L0] The AI subsystem may propose, classify, explain, and escalate. It may not independently authorise a state change. A4 authority is deterministic policy automation, not AI autonomy [see A4].
C-2 7	Non-Marketplace Bound for Add-Ons	[L0] All add-ons in the twenty-eight-add-on catalogue [see Part 5] are non-marketplace bounded. No add-on introduces a public matching engine or anonymous counterparty introduction.

Layer 0 | Constitution

#	Principle	Statement
C-28	Data-Residency Respecting	[L0] SGTX respects the data-residency requirements of each jurisdiction in which it operates. Data elements are classified by residency class [see Appendix B]. No data element is stored or replicated in violation of its residency class.
C-29	Regulatory Classification Respected	[L0] SGTX does not self-classify its regulatory status. Regulatory classification is performed by the Regulatory Classification Gate [see Part 7] per the relevant jurisdiction. Where a jurisdiction requires licensing, SGTX operates only after the licence is granted.
C-30	No Stablecoin Issuance	[L0] SGTX does not issue stablecoins. References to stablecoin-like instruments in earlier documents refer to USTN-backed reserve instruments held at licensed banks, not to SGTX-issued cryptographic tokens on a public blockchain.
C-31	No DeFi Integration	[L0] SGTX does not integrate with decentralised finance protocols. Settlement is bank-authoritative; reserves are held at licensed institutions; no SGTX component touches a public DeFi contract.
C-32	Constitutional Supremacy	[L0] This constitution (Layer 0) is the supreme normative authority within SGTX. Where any lower-layer statement appears to conflict with an L0 statement, the L0 statement governs. The amendment process for this constitution is specified below [see Amendment Process].

The thirty-two points above are mutually reinforcing. C-32 (Constitutional Supremacy) establishes the precedence rule; C-10 through C-13 establish the enforcement mechanisms (Governor, OPA, WasmEdge, Loom); C-01 through C-08 establish the institutional boundary (what SGTX is not); C-09 and C-26 establish the AI boundary; C-14, C-15, and C-20 establish the USTN discipline; C-17 establishes the reserve rule; C-18, C-19, C-23, and C-24 establish the closure and recovery discipline; C-21 and C-30 establish the settlement discipline; C-27 establishes the add-on boundary; C-28 and C-29 establish the regulatory discipline.

AI Authority Ladder (A0 through A5)

The AI Authority Ladder establishes the maximum authority that any AI subsystem may exercise in any context. The ladder is cumulative: a higher rung assumes the lower rungs are also in force. The ladder is specified at L0 because it is a constitutional constraint, not a configuration option. No add-on, no jurisdiction adapter, and no operator may grant AI authority beyond what the ladder permits.

Run g	Authority Level	Permitted AI Actions	Forbidden AI Actions
A0	None	No AI involvement. The operation is performed entirely by deterministic code paths with no AI inference call.	Any AI call, including classification or explanation, is forbidden.
A1	Advisory	AI may propose, classify, explain, and recommend. All AI outputs are presented to a human decision-maker who may accept, modify, or reject them.	AI may not execute. AI may not block execution. AI output has no binding force.
A2	Constraining	AI may flag an operation as out-of-policy, requiring the operation to pass additional review before execution. The flag itself does not block; it narrows the path.	AI may not finally deny. AI may not modify the operation. AI may not execute on its own constraint.
A3	Escalation	AI may escalate an operation to a human or to a higher-authority deterministic policy. Escalation pauses the operation pending resolution.	AI may not resolve its own escalation. AI may not execute as a resolution. AI may not self-promote beyond A3.

Layer 0 | Constitution

Run g	Authority Level	Permitted AI Actions	Forbidden AI Actions
A4	Execution Within Bounds	A deterministic policy engine (OPA + WasmEdge) may execute the operation within pre-authorised, constitutionally-valid bounds. This is deterministic policy automation, not AI autonomy. The AI subsystem may prepare the inputs but does not authorise the execution.	AI may not exceed the pre-authorised bounds. AI may not modify the bounds. AI may not execute outside the bounds.
A5	FORBIDDEN	No AI subsystem may exercise independent execution authority. This rung exists to make explicit that there is no rung above A4.	Any AI execution not bounded by A4 is a constitutional violation and must be rejected by WasmEdge.

A4 is not AI autonomy

Finding A-02 of the v13.0 audit clarified that the phrase *A4 Governance / auto-executes within constitutional bounds* refers to deterministic policy automation executed by the Governor (G2 OPA, G3 WasmEdge) under pre-authorised rules. The AI subsystem itself never acquires independent execution authority at A4; it prepares inputs, and the deterministic engine executes within bounds. This preserves the A4 low-risk automation capability while making the constitutional boundary explicit and auditable.

Non-Custody Principle

[L0] SGTX is non-custodial by architecture. This is an architectural property, not a legal classification. SGTX does not hold customer funds, customer goods, customer documents, or customer title at any point in the transaction lifecycle. The non-custody property is enforced structurally: there is no SGTX-internal account that can hold customer funds, no SGTX-internal warehouse that can hold customer goods, and no SGTX-internal title registry that can hold customer title.

The non-custody property has three operational implications. First, fiat settlement is executed by licensed banks on each payment leg; SGTX orchestrates the settlement instructions but does not itself move fiat. Second, goods movement is executed by licensed carriers; SGTX orchestrates transport instructions but does not itself move goods. Third, document authority remains with the issuing institutions (customs authorities, carriers, banks, inspection agencies); SGTX records document references and verifies their provenance but does not itself issue authoritative documents.

The distinction between architectural property and legal classification is important. SGTX's non-custody is structural: it cannot hold funds because there is no mechanism to hold funds. Whether this structural property results in a particular legal classification (e.g., as a non-financial intermediary) in a particular jurisdiction is a matter for the Regulatory Classification Gate [see Part 7] and the relevant jurisdiction's regulator. SGTX does not pre-empt that classification.

Finding A-03 — Non-Custody vs Payment Functionality

The v13.0 audit clarified that non-custody is compatible with payment/settlement functionality, provided the payment/settlement is executed by licensed institutions and SGTX only orchestrates instructions. Non-custody is not the absence of payment functionality; it is the absence of SGTX-internal fund holding.

Non-Marketplace Principle

[L0] SGTX is not a marketplace. SGTX does not operate a public matching engine. Trade relationships are relationship-controlled: a trade exists between two named, authenticated counterparties who have an established commercial relationship. SGTX does not introduce counterparties, does not rank counterparties for selection, and does not facilitate anonymous trade.

The non-marketplace property is enforced structurally. There is no public order book in SGTX. There is no public listing of buy-side or sell-side intents. A trade request in SGTX names both counterparties at the time of request creation; there is no phase in which a trade request is broadcast to a pool of unknown potential counterparties. SGTX may, however, support relationship-controlled networks (e.g., a buyer and its approved supplier list) within which trade requests may be issued to a named set of pre-approved counterparties; such networks are themselves governed artefacts.

Finding A-16 — Marketplace Terminology

The v13.0 audit clarified that earlier documents occasionally used the term *marketplace* in a colloquial sense (a place where commerce occurs). v14.0 eliminates this colloquial usage entirely. SGTX is a trade execution infrastructure, not a marketplace. The term *marketplace* is reserved for its narrow institutional meaning: a public matching engine. SGTX does not operate one.

The non-marketplace principle extends to all add-ons in the twenty-eight-add-on catalogue [see Part 5]. Every add-on is non-marketplace bounded: no add-on introduces a public matching engine or anonymous counterparty introduction. This is captured in the add-on status matrix by the *Non-Marketplace Bound* column, which is true for all twenty-eight add-ons.

Constitutional Amendment Process

[L0] The Layer 0 constitution may be amended only through the formal amendment process specified below. No other process — including operator action, governance authority action, or technical configuration change — may amend any L0 statement. The amendment process is itself an L0 statement (C-32: Constitutional Supremacy implies that the amendment process is part of the constitution).

Step 1 — Amendment Proposal

An amendment proposal may be initiated by any constitutional signatory. The proposal is recorded as a governed artefact in the event spine with full provenance: proposer identity, proposed text, affected L0 statements, rationale, and impact assessment.

Step 2 — Notice Period

Once recorded, the proposal enters a notice period of no fewer than thirty (30) calendar days. During the notice period, the proposal is visible to all constitutional signatories and to the Loom audit log. No action may be taken on the proposal during the notice period; the notice period exists to permit review and dissent.

Step 3 — Multisig Authorisation

After the notice period expires, the proposal is submitted for multisig authorisation. Constitutional amendments require a 3-of-5 multisig of constitutional signatories [see Appendix C for the multisig policy]. Standard operations (non-constitutional) require 2-of-3 multisig; constitutional amendments require the higher threshold.

Step 4 — WasmEdge Module Update

Once the multisig is obtained, the corresponding WasmEdge module(s) encoding the affected L0 statements are updated. The update is itself a governed operation: the previous module version is preserved, the new module version is signed, and the signature is recorded in the Loom audit log. The amendment takes effect at the moment the new WasmEdge module is signed and loaded.

Step 5 — Audit Trail Preservation

The amendment is recorded in the Loom audit log with full provenance: the proposal, the notice period, the multisig signatures, the previous module version, the new module version, and the timestamp. The previous L0 text remains accessible (the audit trail is append-only); the new L0 text is the operative text from the moment of signing.

No Retrospective Amendment

[L0] Amendments operate prospectively only. An amendment cannot recharacterise a past operation; the past operation remains governed by the L0 text in force at the time of the operation. This is a direct consequence of the Loom audit log's append-only property and the recovery-is-not-erasure principle (C-19).

LAYER 1

Architecture

The current normative architecture of the SGTX platform. Statements in this layer are tagged [L1]. They describe what the platform is, not what it might become. Architectural evolution is governed by Layer 0.

Part 1 — Multi-Clock State Vector Model

[L1] The SGTX platform models every trade as a state vector composed of twelve domain clocks, each advancing independently through six finality classes. The state vector is the canonical view of a trade; all queries against trade state resolve to a state-vector read. The multi-clock model recognises that real-world trades do not advance in lock-step: the commercial leg may be settled while the customs leg is still pending; the documentation leg may be complete while the insurance leg is still being negotiated.

The Twelve Domain Clocks

#	Domain Clock	Scope	Example Finality Milestones
1	Commercial	Contract, pricing, terms, amendments, trade-level commercial milestones.	Contract locked; price amendment; trade acceptance.
2	Logistics	Physical movement: pickup, in-transit, arrival, delivery.	Pickup confirmed; in-transit; arrival at port; delivery accepted.
3	Customs	Customs declarations, clearance, duties, holds, releases.	Declaration filed; clearance granted; duty paid; release.
4	Financial	Payment legs, settlement instructions, settlement confirmations.	Payment leg proposed; instruction signed; settlement confirmed.
5	Documentation	Issuance and verification of trade documents (LC, BL, invoices, certificates).	Document issued; document verified; document transmitted.
6	Compliance	Sanctions, AML, KYC, export controls, trade-restriction screening.	Screening passed; alert raised; review completed.
7	Insurance	Cargo insurance, liability insurance, policy binding, claims.	Policy bound; coverage active; claim filed; claim settled.
8	Quality Control (QC)	Inspection, sampling, certification, conformity assessment.	Inspection scheduled; inspection performed; certificate issued.
9	Dispute	Dispute opening, mediation, arbitration, resolution.	Dispute opened; mediation requested; resolution recorded.
10	Post-Trade	Post-delivery adjustments, returns, warranties, claims.	Warranty active; return requested; adjustment posted.
11	Evidence	Evidence package assembly, sealing, supplementation.	Evidence assembled; evidence sealed; supplementation filed.
12	Governance	Governor decisions, multisig actions, constitutional amendments.	Decision recorded; multisig obtained; amendment signed.

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Finality Classes (F0 through F5)

[L1] Each domain clock advances through six finality classes. The finality class of a clock domain represents the strength of the commitment that has been recorded for that domain. Finality is monotonic within a domain: once a domain has reached F3 (Confirmed), it cannot regress to F2 (Asserted) without an explicit recovery event that is itself recorded in the event spine.

Class	Name	Meaning	Authority Required to Advance
F0	None	No commitment recorded. The domain has not been initiated for this trade.	N/A (initial state).
F1	Proposed	A commitment has been proposed but not yet asserted by an authoritative party.	Any authenticated party may propose; no authority required.
F2	Asserted	An authoritative party has asserted the commitment. The assertion is recorded but not yet confirmed by an external or independent source.	Authoritative party (per domain). E.g., for Financial: the originating bank.
F3	Confirmed	The commitment has been confirmed by an external or independent source. The assertion is now backed by corroborating evidence.	External or independent source (per domain). E.g., for Customs: the customs authority; for Financial: the executing bank.
F4	Settled	The commitment has been settled: the underlying obligation has been discharged. Settlement is irrevocable except through explicit recovery.	Settlement authority (per domain). E.g., for Financial: the executing bank; for Logistics: the carrier and consignee.
F5	Sealed	The commitment has been sealed as part of trade closure. Sealed commitments are immutable except through constitutional amendment or express post-closure observation-period actions [see C-23].	Multisig per the multisig policy [see Appendix C].

Divergence Index

[L1] The Divergence Index is a computed measure of how far the twelve domain clocks have diverged from one another. A trade with all twelve clocks at the same finality class has a divergence index of NONE. A trade where one clock has advanced significantly beyond the others has a higher divergence index. The index is used by the Governor to trigger additional review when a trade is materially out of balance.

Index	Meaning	Governor Action
NONE	All clocks within one finality class of each other.	No action; routine progression.
LOW	One or two clocks ahead by two finality classes.	Continue; flag for routine monitoring.
MEDIUM	One clock ahead by three or more finality classes, or three or more clocks ahead by two classes.	Require explicit Governor review at next state-changing operation.
HIGH	Multiple clocks ahead by three or more classes, or any single clock at F4 while others remain at F1 or below.	Block next state-changing operation pending human review; raise SEV-2 incident [see Appendix C].
CRITICAL	Any clock at F5 (Sealed) while another clock remains at F1 or below, indicating attempted closure with material unconfirmed commitments.	Block all operations on the trade; raise SEV-1 incident; require constitutional signatory review.

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State Integrity Computation

[L1] The integrity of a trade's state vector is computed as the composition of three properties: (i) monotonicity within each clock domain (no regressions without explicit recovery events); (ii) hash chain integrity of the event spine that drives the state vector [see Part 2]; and (iii) closure-policy compatibility [see Part 8]. A trade state vector is *integral* if and only if all three properties hold. The Governor refuses to act on a non-integral state vector; recovery is mandatory before any further state change.

Part 2 — Event Spine (Immutable Event Log)

[L1] The event spine is the immutable, hash-chained record of every event that has occurred in the platform. Every state change is reflected as one or more events on the spine. The spine is the authoritative history; the state vector [see Part 1] is a derived view rebuilt from the spine. A trade's entire history can be reconstructed by replaying its events from the spine.

Command / Event Taxonomy

[L1] Finding A-20 of the v13.0 audit established the formal separation of commands from events. v14.0 elevates this to a first-class architectural principle. A **command** is an instruction that requests a state change. An **event** is an immutable record that a state change occurred (or did not occur, with reasons). The distinction is captured in the table below.

Property	Command	Event
Mutability	Mutable; may be modified, withdrawn, or rejected.	Immutable; once recorded, never modified or deleted.
Authority	Issued by an authenticated party.	Recorded by the Governor with full provenance.
Effect	Requests a state change; has no direct state effect.	Records a state change (or its rejection); is the state record.
Replay	May be replayed; replay may produce a different outcome if conditions have changed.	Replay reproduces the exact historical state; the spine is deterministic.
Storage	Stored in the command queue; periodically pruned.	Stored in the event spine; never pruned.
Hash Chain	Not hash-chained.	Hash-chained: each event's hash incorporates the previous event's hash.

Event Taxonomy — Observation / Assertion / Confirmation

[L1] Events are classified into three authority categories, each carrying different authority attributes. The category determines how the event contributes to finality advancement [see Part 1, F0-F5].

Category	Authority	Finality Impact
Observation	Recorded by an SGTX component that observed an external state but did not cause it. E.g., a logistics tracker observing a vessel arrival from a carrier API.	Advances the relevant clock to F1 (Proposed) at most; cannot advance beyond F1 without corroboration.
Assertion	Recorded by an authoritative party that asserts a state. E.g., a carrier asserting pickup; a bank asserting a payment instruction.	Advances the relevant clock to F2 (Asserted); cannot advance to F3 without confirmation.
Confirmation	Recorded by an external or independent source that confirms a previously asserted state. E.g., a customs authority confirming clearance; an executing bank confirming settlement.	Advances the relevant clock to F3 (Confirmed) or beyond, depending on the nature of the confirmation.

SHA-256 Hash Chain

[L1] Each event in the spine carries a SHA-256 hash that incorporates the event's content and the hash of the previous event. The hash chain ensures that any modification to a historical event would invalidate the hashes of all subsequent events, making tampering detectable. The hash chain is replicated to a quorum of audit nodes; the quorum protocol is specified in Layer 2, Part 14 (Observability Catalogue) [see Part 14].

Event Types Catalogue

[L1] The platform defines fifty or more canonical event types. The full catalogue is specified in Layer 2, Part 11 (Canonical Event Type Catalogue) [see Part 11]. Each event type defines its required fields, its authority class (Observation / Assertion / Confirmation), its clock impact (which domain clock it advances and to what finality), and its lineage rules (which prior events it depends on and which subsequent events it permits).

Replay Mode (Section 86)

[L1] The event spine supports replay mode, referenced in the source documents as Section 86. In replay mode, the spine is read in sequence and the state vector is reconstructed event-by-event. Replay mode is used for: (i) state reconstruction after a node restart; (ii) audit verification by external auditors; (iii) dispute resolution, where the exact sequence of events must be demonstrated; and (iv) recovery, where a failed state is reconstructed to determine the recovery path. Replay is deterministic: the same spine always produces the same state vector.

Part 3 — Governor Pipeline

[L1] The Governor is the constitutional enforcement component of the SGTx platform. Every state-changing operation passes through the seven-gate Governor pipeline before its effect is committed. The seven gates are defined as constitutional principles at L0 (G1 through G7) [see L0 Governor Principles]; this Part specifies their architectural realisation.

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The Seven Gates (Architectural Realisation)

Gat e	L0 Principle	Architectural Component
G1	Execution Always Gated	The Governor pipeline is the single entry point for state-changing operations. There is no bypass path; there is no administrative override path. The pipeline is implemented as a sequenced middleware that wraps every command.
G2	OPA Enforced	All policy decisions are evaluated by Open Policy Agent (OPA) using Rego modules stored in a versioned policy repository. The Rego modules are themselves governed artefacts: changes are signed, versioned, and recorded in the Loom audit log.
G3	WasmEdge Constitutional	The L0 constitutional constraints are encoded as WasmEdge modules. WasmEdge executes as a hard boundary around OPA: it receives the OPA decision and the command context, and may veto the decision if it violates the constitution. The WasmEdge modules are signed and versioned; updates require constitutional amendment [see L0 Amendment Process].
G4	Loom Audited	Every Governor decision (ALLOW, DENY, CONDITIONAL) and every committed state change is recorded in the Loom audit log. Loom is an append-only, hash-chained, quorum-replicated log. No operation may execute without a corresponding Loom entry; the Loom entry is written before the state change is committed.
G5	Multisig for Irreversible	Operations classified as irreversible (settlement release, USTN sealing, constitutional amendment, evidence destruction) require multisig authorisation per the multisig policy [see Appendix C]. The multisig check is performed by the Governor before the operation is committed.
G6	AI Advisory Only	The AI subsystem may propose, classify, and escalate. Its outputs are inputs to the Governor; they are not decisions. The Governor records AI outputs alongside its decision, but the decision itself is made by the deterministic OPA + WasmEdge pipeline.
G7	Bank-Authoritative Settlement	For settlement operations, the Governor verifies that the executing party is the licensed bank on the relevant payment leg. SGTX does not execute fiat movements; the bank does. The Governor records the bank's settlement confirmation as an event.

Decision Merge Semantics

[L1] The seven gates produce independent decisions. The merge semantics determine the final Governor decision from the seven gate decisions. The merge order is strict: DENY overrides all; CONDITIONAL narrows; ALLOW is permissive only in the absence of DENY or CONDITIONAL.

Decision	Precedence	Effect
DENY	Highest	The operation is rejected. The rejection is recorded in the Loom audit log with the denying gate and the reason. No state change occurs.
CONDITIONAL	Middle	The operation is permitted only within the conditions specified by the conditioning gate. The conditions become part of the operation context; subsequent gates evaluate against the conditioned operation.
ALLOW	Lowest	The operation is permitted by this gate. If no other gate DENYs or CONDITIONALS, the operation proceeds.

The merge result is therefore: **DENY** if any gate DENYs; **CONDITIONAL** if no gate DENYs but at least one gate CONDITIONALS (the conditions are the union of all conditioning gates' conditions); **ALLOW** if all gates ALLOW. This is summarised as *DENY > CONDITIONAL > ALLOW*.

Governor Decision Lifecycle

[L1] A Governor decision progresses through four phases: (i) **intake**, where the command is received and the operation context is assembled; (ii) **gate evaluation**, where each of the seven gates evaluates independently; (iii) **merge**, where the seven decisions are merged per the semantics above; and (iv) **commit**, where the merged decision is recorded in the Loom audit log and, if ALLOW or CONDITIONAL, the state change is committed to the event spine. The commit phase is atomic: either the Loom entry and the event spine entry are both written, or neither is.

Part 4 — Settlement Orchestration Control Plane

[L1] The Settlement Orchestration Control Plane is the architectural component that orchestrates the multi-leg settlement of a trade. SGTX orchestrates; banks execute. SGTX does not hold funds (per the Non-Custody Principle, L0) and does not execute fiat movements (per G7, Bank-Authoritative Settlement).

Multi-Leg Payment Model

[L1] A trade may have multiple payment legs. Each payment leg is modelled as a **PaymentLeg** with a payer, payee, currency, amount, execution window, and atomicity policy. A **SettlementInstruction** groups one or more PaymentLegs that must be settled together under a specified atomicity policy. The instruction is signed by the originating party and submitted to the Governor for validation before being sent to the relevant bank(s) for execution.

Entity	Fields	Authority
PaymentLeg	leg_id, payer_party_id, payee_party_id, currency, amount, execution_window_start, execution_window_end, atomicity_policy, bank_id, purpose_code, regulatory_references.	Issued by the originating party; validated by the Governor; executed by the named bank.
SettlementInstruction	instruction_id, list of PaymentLeg_ids, atomicity_policy, originator_signature, governor_decision_ref, submission_timestamp.	Issued by the originating party; validated by the Governor; submitted to banks for execution.
SettlementConfirmation	confirmation_id, instruction_id, leg_id, executing_bank_id, bank_transaction_ref, executed_amount, executed_timestamp, confirmation_signature.	Issued by the executing bank; recorded as a Confirmation event on the event spine.

Atomicity Policies

[L1] Five atomicity policies govern how a SettlementInstruction's PaymentLegs are settled. The policy is set at instruction creation and may not be modified after submission.

Policy	Meaning	Failure Behaviour
ALL_OR_NONE	All legs must settle successfully; any leg failure causes the entire instruction to be rolled back.	On any leg failure: previously settled legs are reversed (where possible) or marked for reconciliation.
PARTIAL_ALLOWED	Each leg settles independently; partial settlement is permitted.	On leg failure: the failed leg is marked failed; other legs continue.
SEQUENCED	Legs settle in a specified sequence; each leg's success is a precondition for the next leg's execution.	On leg failure: subsequent legs are not executed; previously settled legs remain settled.

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Policy	Meaning	Failure Behaviour
CONDITIONAL	Legs settle subject to conditions specified in the instruction (e.g., customs clearance confirmation, delivery acceptance).	On condition failure: the leg is held pending; if the condition is not met within the execution window, the leg fails.
HUMAN_RELEASE	Legs are held pending explicit human release. Used for high-value or high-risk settlements.	On release: the leg executes; on refusal: the leg is cancelled.

Bank Settlement Gateway — Six-Stage Pipeline

[L1] The Bank Settlement Gateway is the component that interfaces with the executing bank's payment infrastructure. Each SettlementInstruction passes through a six-stage pipeline before the bank executes the fiat movement. The pipeline is implemented per bank connector; each connector is a governed artefact.

Stage	Name	Purpose
1	Schema Validation	Validate the instruction against the bank's schema (message format, field presence, field formats).
2	Signature Verification	Verify the originator's signature and the Governor decision reference. Reject if either is invalid or missing.
3	USTN Resolution	Resolve the USTN namespace to verify the trade context. The USTN is a namespace, not an override of bank authority [see C-20].
4	Beneficiary Verification	Verify the payee against the bank's beneficiary records. Reject if the beneficiary is unknown, sanctioned, or otherwise ineligible.
5	Bank Policy	Apply the bank's internal policy (limits, corridors, currency controls, cut-off times). The bank may reject or hold the instruction per its own policy.
6	AML / Sanctions Screening	Screen the payer, payee, and purpose code against AML and sanctions lists. Reject if any party is flagged.

Non-Custody Boundary

Non-Custody Boundary

[L1] SGTX orchestrates settlement instructions; it never holds funds. The payment legs are executed by the licensed banks on each leg. At no point in the settlement lifecycle does SGTX itself receive, hold, or transfer fiat currency. The SettlementInstruction is an instruction; the SettlementConfirmation is a record of the bank's execution. Neither involves SGTX custody. This is the architectural realisation of the Non-Custody Principle [see L0 Non-Custody] and the Bank-Authoritative Settlement gate (G7).

Part 5 — 28-Add-On Catalogue with Status Matrix

[L1] The SGTX platform defines a canonical catalogue of twenty-eight add-ons. Each add-on extends the platform with a specific capability. All add-ons are non-marketplace bounded (per C-27): no add-on introduces a public matching engine or anonymous counterparty introduction. Add-ons are governed by the same constitutional principles (L0) and architectural commitments (L1) as the core platform; an add-on cannot override or weaken any L0 or L1 commitment.

Status Matrix

[L1] Each add-on carries a status indicating its current deployment state. The status vocabulary is established by finding A-19 of the v13.0 audit and formalised in v14.0. The four status values are:

Status	Meaning
CORE_READY	The internal architecture is specified, implemented, and verified against the L2 implementation specifications.
PRODUCTION_CONNECTED	The add-on is wired to a real, live external system and round-trip transactions have been observed in production-equivalent environments.
LEGAL_AUTHORIZATION_REQUIRED	The add-on requires explicit legal authorisation (licence, registration, regulatory clearance) per the relevant jurisdiction before it can be activated for production traffic.
FOUNDATION	The add-on depends on foundational capabilities that are themselves in CORE_READY or PRODUCTION_CONNECTED state; the add-on cannot advance beyond its current state until its dependencies advance.

[L1] The full status matrix for the twenty-eight-add-on catalogue is below. Priority is the implementation priority per the priority framework [see Appendix D]. Blueprint Reference is the source document section that specifies the add-on in detail. Non-Marketplace Bound is true for all twenty-eight add-ons, reflecting the constitutional constraint (C-27).

#	Add-On Name	Priority	Status	BP Ref	NMB
1	GNN Risk Engine	P2	CORE_READY	AO-01	Yes
2	Federated Learning	P2	CORE_READY	AO-02	Yes
3	Causal Inference Engine	P2	CORE_READY	AO-03	Yes
4	Self-Healing Infrastructure	P2	CORE_READY	AO-04	Yes
5	Automated Penetration Testing	P2	CORE_READY	AO-05	Yes
6	Post-Quantum Cryptography	P2	CORE_READY	AO-06	Yes
7	Expanded ZK Proofs and Proof of Reserves	P2	CORE_READY	AO-07	Yes
8	Customs Bond and Guarantee Management	P2	LEGAL_AUTHORIZATION_REQUIRED	AO-08	Yes
9	Demurrage and Detention	P3	CORE_READY	AO-09	Yes
10	Broker Liability and Insurance	P3	CORE_READY	AO-10	Yes
11	Customs Valuation Intelligence	P3	CORE_READY	AO-11	Yes
12	Cold Chain Quality Management	P3	CORE_READY	AO-12	Yes
13	Inspection Agency Accreditation	P3	CORE_READY	AO-13	Yes

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#	Add-On Name	Priorit y	Status	BP Ref	NMB
14	Currency Risk Management	P3	LEGAL_AUTHORIZATION_REQUIRED	AO-14	Yes
15	Government API Sandbox	P3	CORE_READY	AO-15	Yes
16	FTA Preference Management	P3	CORE_READY	AO-16	Yes
17	Piracy and Security Risk Engine	P3	CORE_READY	AO-17	Yes
18	Trade Compliance Calendar	P3	CORE_READY	AO-18	Yes
19	Cargo Insurance Integration	P4	PRODUCTION_CONNECTED	AO-19	Yes
20	Trade Finance Documentation	P4	LEGAL_AUTHORIZATION_REQUIRED	AO-20	Yes
21	Back-to-Back LC Management	P4	LEGAL_AUTHORIZATION_REQUIRED	AO-21	Yes
22	Force Majeure Handling	P4	CORE_READY	AO-22	Yes
23	Shipper's Declaration and Export Docs	P4	CORE_READY	AO-23	Yes
24	Port and Terminal Integration	P4	PRODUCTION_CONNECTED	AO-24	Yes
25	Payment Guarantee Confirmation (Optional)	P4	LEGAL_AUTHORIZATION_REQUIRED	AO-25	Yes
26	Demurrage Dispute Resolution	P4	CORE_READY	AO-26	Yes
27	(RESERVED)	P4	FOUNDATION	AO-27	Yes
28	GRiRE (Global Regulatory Intelligence and Requirements Engine)	P4	CORE_READY	AO-28	Yes

Add-On Detailed Summaries

[L1] The following subsections provide a brief summary of each add-on. Each summary identifies the add-on's scope, its non-marketplace boundary, and its key dependencies. Full specifications are preserved in the source documents.

Add-On 1 — GNN Risk Engine (non-marketplace bounded)

A Graph Neural Network-based risk engine that evaluates trade risk across counterparties, geographies, and commodities. Operates on the relationship-controlled trade graph; does not introduce anonymous matching. Outputs are advisory (A1) and feed into the Governor as inputs.

Add-On 2 — Federated Learning

A federated learning framework that enables model training across tenant data without centralising the data. Preserves data residency (per C-28) and tenant isolation. Models trained federated are versioned and governed.

Add-On 3 — Causal Inference Engine

A causal inference engine that distinguishes correlation from causation in trade analytics. Used for post-trade analysis and risk attribution. Outputs are advisory (A1).

Add-On 4 — Self-Healing Infrastructure

An infrastructure layer that detects and recovers from component failures automatically. Recovery actions are themselves governed: they pass through the Governor pipeline. Recovery is not erasure (per C-19); the failed path is preserved in the event spine.

Add-On 5 — Automated Penetration Testing

A continuous automated penetration testing capability that probes the platform for vulnerabilities. Findings are recorded as governed artefacts; remediation passes through the change-management process [see Appendix C].

Add-On 6 — Post-Quantum Cryptography

A cryptographic layer that uses post-quantum algorithms for signatures and key exchange. Ensures forward secrecy against quantum-capable adversaries. Applied to all hash-chain signatures and settlement instruction signatures.

Add-On 7 — Expanded ZK Proofs and Proof of Reserves

A zero-knowledge proof system that provides cryptographic attestation of the 110% reserve rule (per C-17) without revealing reserve composition details. The proof is published; the underlying composition is attested but not disclosed in detail. Specified in Appendix A [see Appendix A].

Add-On 8 — Customs Bond and Guarantee Management

Manages customs bonds and guarantees required by customs authorities. Requires legal authorisation per jurisdiction; the add-on cannot operate without the relevant customs authority's authorisation. Non-marketplace bounded: bonds are issued per established relationships, not publicly matched.

Add-On 9 — Demurrage and Detention

Calculates and tracks demurrage and detention charges for container storage at ports. Integrates with port and terminal systems (Add-On 24). Outputs feed into settlement instructions where demurrage is owed.

Add-On 10 — Broker Liability and Insurance

Manages liability and insurance for customs brokers and freight forwarders. Tracks policy binding, coverage, and claims. Non-marketplace bounded: insurance is per established counterparty-insurer relationships.

Add-On 11 — Customs Valuation Intelligence

Provides intelligence on customs valuation methods per jurisdiction. Used to validate declared values against jurisdiction-specific rules. Outputs are advisory (A1) and feed into the customs declaration workflow.

Add-On 12 — Cold Chain Quality Management

Manages quality for cold-chain shipments, including temperature monitoring, excursion detection, and chain-of-custody verification. Integrates with IoT sensors and carrier systems.

Add-On 13 — Inspection Agency Accreditation

Tracks inspection agency accreditation status per jurisdiction. Inspection certificates are accepted only from accredited agencies. Non-marketplace bounded: inspections are performed by named, accredited agencies.

Add-On 14 — Currency Risk Management

Provides currency hedging instruments for trade settlement. Requires legal authorisation per jurisdiction (derivatives licensing). Non-marketplace bounded: hedging is per established counterparty-bank relationships.

Add-On 15 — Government API Sandbox

A sandbox environment for testing government API integrations without affecting production government systems. Used for connector development and certification. The sandbox is governed; sandbox operations are recorded in the Loom audit log.

Add-On 16 — FTA Preference Management

Manages Free Trade Agreement preference claims, including rules of origin verification and certificate of origin management. Non-marketplace bounded: preferences are claimed per established trade relationships.

Add-On 17 — Piracy and Security Risk Engine

Evaluates piracy and security risks for ocean shipments, including high-risk area routing and insurance implications. Outputs are advisory (A1) and feed into route planning.

Add-On 18 — Trade Compliance Calendar

Maintains a calendar of trade compliance deadlines per trade and per jurisdiction. Triggers alerts and Governor reviews when deadlines approach.

Add-On 19 — Cargo Insurance Integration

Integrates with cargo insurance providers to bind policies, file claims, and track settlements. Non-marketplace bounded: insurance is per established counterparty-insurer relationships. Status: PRODUCTION_CONNECTED — live with selected insurers.

Add-On 20 — Trade Finance Documentation

Manages trade finance documentation including LCs, guarantees, and standby LCs. Requires legal authorisation per jurisdiction. Non-marketplace bounded: documentation is per established counterparty-bank relationships.

Add-On 21 — Back-to-Back LC Management

Manages back-to-back letter of credit arrangements, where an LC is opened against an existing LC. Requires legal authorisation per jurisdiction. Non-marketplace bounded.

Add-On 22 — Force Majeure Handling

Handles force majeure declarations and their impact on trade obligations. Records force majeure events on the event spine; adjusts obligation timing per the contract terms.

Add-On 23 — Shipper's Declaration and Export Docs

Manages shipper's declarations of origin and other export documentation. Integrates with customs export systems. Non-marketplace bounded: declarations are per established trade relationships.

Add-On 24 — Port and Terminal Integration

Integrates with port and terminal operating systems for container tracking, gate movements, and availability. Status: PRODUCTION_CONNECTED — live with selected ports.

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Add-On 25 — Payment Guarantee Confirmation (Optional)

An optional add-on that confirms payment guarantees from third-party guarantors. Requires legal authorisation per jurisdiction. Non-marketplace bounded.

Add-On 26 — Demurrage Dispute Resolution

Provides a workflow for resolving demurrage disputes between shippers, consignees, and terminals. Records dispute events on the event spine; routes to mediation or arbitration per contract terms.

Add-On 27 — (RESERVED)

Reserved for future allocation. No capability is currently specified for this add-on slot. The reservation preserves numbering stability for future additions.

Add-On 28 — GRiRE (Global Regulatory Intelligence and Requirements Engine)

A global regulatory intelligence engine that tracks regulatory changes across jurisdictions and updates the jurisdiction capability adapters [see Part 6] accordingly. Outputs are advisory (A1) for regulatory changes and authoritative (via the Governor) for adapter updates.

Part 6 — Jurisdiction Capability Adapter Schema

[L1] The Jurisdiction Capability Adapter Schema is the architectural component that enables SGTX to operate across multiple jurisdictions. Each jurisdiction has its own profile, capability matrix, authority map, evidence profile, and payment rail profile. Adapters are the extensibility surface for jurisdiction-specific integrations.

Country Profile, Jurisdiction Profile, Capability Matrix

Entity	Scope	Key Fields
Country Profile	Country-level metadata: ISO codes, currency, timezone, official languages, primary regulator, data-residency classification.	iso_3166_1_alpha2, iso_3166_1_alpha3, currency_code, primary_regulator_id, data_residency_default.
Jurisdiction Profile	A jurisdiction within a country (e.g., a free trade zone, a special economic zone, or a sub-national regulatory region).	jurisdiction_id, parent_country_id, jurisdiction_type, applicable_regulations[], authority_map_id.
Capability Matrix	A per-jurisdiction, per-capability matrix indicating whether each SGTX capability is available, restricted, or forbidden in that jurisdiction.	jurisdiction_id, capability_id, status (AVAILABLE / RESTRICTED / FORBIDDEN), conditions[], legal_basis.

Authority Map, Evidence Profile, Payment Rail Profile

Profile	Purpose
Authority Map	Identifies the authoritative party for each operation type in the jurisdiction. E.g., for customs clearance in Egypt: the Egyptian Customs Authority; for settlement in Egyptian pounds: the licensed Egyptian bank; for cargo release: the port authority.
Evidence Profile	Specifies the evidence requirements for each operation type. E.g., for customs clearance: the customs entry number, the supporting documents (invoice, packing list, certificate of origin), and the customs broker licence.
Payment Rail Profile	Specifies the payment rails available in the jurisdiction, including the central bank rail (e.g., RTGS), the retail rail (e.g., ACH-equivalent), and any intra-bank rails. Identifies settlement cut-off times and currency controls.

Layer 1 | Architecture

Connector States (16 states)

[L1] Each jurisdiction adapter connector progresses through sixteen states, from initial discovery through production connection to eventual deprecation. The states are organised into four lifecycle phases: discovery, development, production, and deprecation.

Phase	States
Discovery	NOT_DISCOVERED, DISCOVERED, PROFILE_DRAFTED, PROFILE_REVIEWED
Development	ADAPTER_SPECIFIED, ADAPTER_IMPLEMENTED, ADAPTER_TESTED, SANDBOX_CONNECTED, SANDBOX_VERIFIED
Production	STAGING_CONNECTED, STAGING_VERIFIED, PRODUCTION_CONNECTED, PRODUCTION_VERIFIED
Deprecation	DEPRECATION_NOTIFIED, DEPRECATED, RETIRED, ARCHIVED

Authoritative Statuses (7 statuses)

[L1] Separately from the connector lifecycle states, each jurisdiction adapter carries an authoritative status indicating the authoritative standing of the integration in the jurisdiction. The seven statuses are:

Status	Meaning
SGTX_READY	SGTX is technically ready to operate in this jurisdiction.
REGULATOR_NOTIFIED	The relevant regulator has been notified of SGTX's intent to operate.
REGULATOR_REVIEWING	The regulator is reviewing SGTX's filing.
REGULATOR_CONDITIONAL	The regulator has granted conditional approval.
REGULATOR_APPROVED	The regulator has granted full approval.
GOVERNMENT_RELEASED	The relevant government authority has released SGTX for production use.
SUSPENDED	The integration is suspended (by SGTX, the regulator, or the government).

Four-Dimension External Readiness

[L1] Each jurisdiction's external readiness is evaluated along four independent dimensions. No dimension may be inferred from another. A jurisdiction is fully production-ready only when all four dimensions are satisfied.

Dimension	Scope	Evidence
TECHNICAL	The technical integration with government, bank, carrier, and customs systems is operational and verified.	Connector state PRODUCTION_VERIFIED; end-to-end test transactions.
LEGAL	All required legal authorisations are in place (licences, registrations, regulatory clearances).	Authoritative status GOVERNMENT_RELEASED; legal opinion on file.
OPERATIONAL	Operational runbooks, support staff, and incident response procedures are in place for the jurisdiction.	Operational runbook; staff training records; incident drill completion.
COMMERCIAL	Commercial relationships with local banks, carriers, customs brokers, and other required parties are established.	Signed commercial agreements; counterparty onboarding records.

Part 7 — Regulatory Classification Gate

[L1] The Regulatory Classification Gate maps the actual functionality provided by SGTX in a given jurisdiction to the licence class required by that jurisdiction's regulator. The gate is conservative: where functionality is ambiguous, the gate assigns the higher licence class. SGTX operates in a jurisdiction only after the required licence is granted.

Mapping Functionality to Licence Class

[L1] The mapping is performed per jurisdiction, per capability. For each capability that SGTX provides, the gate identifies the corresponding regulatory classification in the jurisdiction and determines the licence class required. The classification is recorded in the Capability Matrix [see Part 6].

Capability	Typical Classification	Licence Required
Settlement orchestration	Payment service provider (varies)	Per jurisdiction; in Egypt: CBE authorisation per Law 194/2020.
Customs declaration filing	Customs broker	Per jurisdiction; in Egypt: customs broker licence.
Cargo insurance binding	Insurance intermediary	Per jurisdiction; insurance intermediary licence.
Trade finance documentation	Banking service	Per jurisdiction; banking licence or partnership with licensed bank.
Data processing	Data controller / processor	Per jurisdiction; data protection registration.

Non-Custody is Architectural, Not Legal Classification

Non-Custody is Architectural

[L1] SGTX's non-custody is an architectural property [see L0 Non-Custody]: there is no SGTX-internal mechanism to hold customer funds. Whether this architectural property results in a particular legal classification (e.g., as a non-financial intermediary, or as a regulated payment service provider despite non-custody) is a matter for the relevant jurisdiction's regulator. SGTX does not pre-empt that classification; the Regulatory Classification Gate records the regulator's determination, not SGTX's preference.

Egypt / CBE Law 194/2020 as Initial Reference

[L1] The initial reference jurisdiction for SGTX is Egypt, with the Central Bank of Egypt (CBE) Law 194/2020 as the initial regulatory reference. The four direct Egypt integrations (Nafeza, CargoX, ETA, CBE) are the first concrete realisations of the worldwide adapter fabric. Finding A-01 of the v13.0 audit clarified that *Direct API* denotes a currently adopted first-party connector, not a limitation on the global adapter model; the worldwide adapter fabric is the canonical extensibility surface.

[L1] The Egypt regulatory classification is detailed in Appendix B (Regulatory and Legal) [see Appendix B]. The classification is extensible: as SGTX enters additional jurisdictions, their classifications are added to the same matrix.

Part 8 — Closure Policy

[L1] Closure of an SGTx trade is earned, not granted (per C-18, Closure-Is-Earned). A trade is CLOSED only when all seven closure conditions are met. Closure is recorded as an event on the event spine and triggers the sealing of evidence packages (per C-24, Evidence Sealing). After closure, a trade enters a post-closure observation period (per C-23, Observation Period) during which certain event types may still be appended.

The Seven Closure Conditions

#	Condition	Domain Clock	Required Finality
1	Delivery accepted by consignee	Logistics	F3 (Confirmed)
2	Settlement complete on all payment legs	Financial	F4 (Settled)
3	Reconciliation complete	Financial	F3 (Confirmed)
4	Customs complete	Customs	F4 (Settled)
5	Post-clearance complete	Customs	F3 (Confirmed)
6	Disputes satisfied	Dispute	F4 (Settled) or no open disputes
7	Evidence sealed	Evidence	F5 (Sealed)

canClose Predicate

[L1] The **canClose** predicate is the deterministic function that evaluates whether all seven closure conditions are met. The predicate is implemented as OPA policy (per G2) and is itself governed by WasmEdge (per G3). The predicate's output is one of CAN_CLOSE, CANNOT_CLOSE (with reasons), or CAN_CLOSE_WITH_EXCEPTION (with exceptions and severities).

CLOSED_WITH_EXCEPTION Severity Matrix

[L1] In certain cases, a trade may be closed with exceptions: specific closure conditions are not fully met, but the trade is nevertheless closed (e.g., a minor documentation gap that does not affect the substantive trade). The CLOSED_WITH_EXCEPTION severity matrix governs which exceptions are permitted and which block closure.

Severity	Meaning	Closure Permitted?
1	Cosmetic / typographical issue with no substantive impact.	Yes — closure permitted with note.
2	Minor documentation gap with documented mitigation.	Yes — closure permitted with mitigation plan.
3	Material documentation gap or unresolved minor dispute.	No — closure blocked; requires resolution or explicit constitutional signatory override.
4	Unresolved material dispute or unconfirmed settlement leg.	No — closure blocked; requires resolution.
5	Regulatory non-compliance or unresolved SEV-1 incident.	No — closure blocked; requires regulator engagement.

Post-Closure Event Rules (Section 22 Observation Period)

[L1] After closure, a trade enters the post-closure observation period (referenced as Section 22 in the source documents). During the observation period, certain event types may still be appended to the event spine:

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- **Dispute reopen:** A dispute may be reopened if new evidence surfaces. The reopen event must reference the original closed dispute and provide the new evidence.
- **Evidence supplementation:** Additional evidence may be appended. The supplementation is recorded as a new evidence package with explicit lineage to the sealed package [see C-24].
- **Audit correction:** An audit finding may be recorded. The correction does not modify the original event; it adds an annotation event with the correction.
- **Regulatory inquiry:** A regulator may query the trade's history. The query is logged; the response is generated from the event spine.

[L1] The observation period does not permit arbitrary modification. Sealed events remain sealed; settled payment legs remain settled. The observation period is a window for■■■ events that build on the closed state, not a window for rewriting the closed state.

Part 9 — AI Recommendation Gateway

[L1] The AI Recommendation Gateway is the architectural component that mediates between the AI subsystem and the Governor. The gateway enforces the AI Authority Ladder [see L0 A0 through A5]: AI proposes, the Governor validates, OPA enforces, WasmEdge performs the constitutional check. AI never has independent execution authority (per C-26 and G6).

Recommendation Flow

[L1] An AI recommendation progresses through four stages:

- 1 **AI proposes.** The AI subsystem generates a recommendation (e.g., classify this trade as high-risk, escalate this payment leg to human review, propose a customs valuation). The proposal includes the recommendation, the inputs considered, the model version, and the confidence score.
- 2 **Governor validates.** The Governor receives the proposal and validates its admissibility: is the AI subsystem authorised to make this kind of recommendation? Is the model version current? Are the inputs within the AI subsystem's permitted data scope?
- 3 **OPA enforces.** The Governor passes the validated proposal to OPA, which evaluates the proposal against the policy. OPA determines whether to accept, modify, or reject the proposal, and what action (if any) to take.
- 4 **WasmEdge constitutional check.** WasmEdge verifies that the OPA decision does not violate the constitution. If the decision is constitutional, it is committed; if not, WasmEdge vetoes.

AI Never Has Independent Execution Authority

AI Authority Boundary

[L1] The AI subsystem may propose, classify, explain, and escalate. It may not execute. The execution authority at A4 is deterministic policy automation (OPA + WasmEdge), not AI autonomy. The AI subsystem prepares inputs; the deterministic engine executes within pre-authorised bounds. This is the architectural realisation of C-26 (AI Advisory Only) and the A4 rung of the AI Authority Ladder.

Part 10 — Transport Engine Architecture

[L1] The Transport Engine Architecture provides the multimodal transport orchestration capability for SGTX. Five transport modes are supported as first-class citizens: Road, Air, Ocean Container, RoRo (Roll-on/Roll-off), and Rail. A Multimodal Orchestrator coordinates sequences of mode-specific legs into a single trade journey. The USTN serves as the universal namespace across all modes (per C-14, USTN-Centric; and C-20, USTN as Namespace Not Override).

Five Transport Modes

Mode	Scope	Mode-Specific Data
Road	Trucking, including full-truckload (FTL), less-than-truckload (LTL), and special-purpose road transport.	Vehicle registration, driver identification, route plan, border crossing points.
Air	Air cargo, including general cargo, special cargo (perishable, dangerous goods), and express.	AWB number, flight schedule, ULD (Unit Load Device) identification, IATA dangerous goods declaration.
Ocean Container	Containerised ocean shipping, including FCL (Full Container Load) and LCL (Less than Container Load).	B/L number, container number, vessel schedule, port pair, carrier SCAC code.
RoRo (Roll-on/Roll-off)	Roll-on/Roll-off shipping for vehicles, heavy equipment, and other rolling stock.	RoRo B/L number, vehicle identification, stowage plan, port pair.
Rail	Rail freight, including container rail, bulk rail, and intermodal rail.	Railway consignment note, wagon identification, route plan, border crossing points.

Finding A-09 — RoRo as First-Class Mode

[L1] The v13.0 audit clarified that RoRo is a first-class transport mode, not a submode of Ocean Container. While both involve ocean shipping, the operational characteristics (rolling cargo vs. containerised cargo), the documentation (RoRo B/L vs. container B/L), and the port infrastructure (RoRo terminals vs. container terminals) are distinct. v14.0 reflects this as five first-class modes, with RoRo elevated to its own mode rather than treated as a submode of Ocean.

Multimodal Orchestrator

[L1] The Multimodal Orchestrator coordinates sequences of mode-specific legs into a single trade journey. A journey may involve multiple modes (e.g., road to port, ocean to destination port, road to consignee). The orchestrator maintains the journey state, the mode-specific leg states, and the handoff points between legs. Each leg is itself a governed sub-trade with its own USTN (a child USTN of the parent trade USTN, per the multi-shipment model).

USTN as Universal Namespace Across All Modes

[L1] The USTN is the universal namespace across all transport modes. A single trade USTN anchors the entire journey; each mode-specific leg carries a child USTN that references the parent. The USTN does not override the mode-specific identifiers (AWB numbers, B/L numbers, railway consignment note numbers); those identifiers are preserved in the External Identifier Registry [see Part 12] and are linked to the USTN. The USTN is the SGTX namespace; the external identifiers are the authoritative identifiers in their respective external systems.

LAYER 2

Implementation Specifications

Canonical engineering specifications. Statements in this layer are tagged [L2]. They are expected to be implemented and verified against; their textual form is intended to be refined as implementation proceeds.

Part 11 — Canonical Event Type Catalogue

[L2] The Canonical Event Type Catalogue defines the complete set of event types that may be recorded on the event spine. Each event type defines its required fields, its authority class (Observation / Assertion / Confirmation per Part 2), its clock impact (which domain clock it advances and to what finality), and its lineage rules (which prior events it depends on and which subsequent events it permits). The catalogue is normative: no event type may be recorded on the spine unless it is defined in this catalogue.

Common Required Fields (All Event Types)

[L2] Every event, regardless of type, carries the following common required fields:

Field	Type	Description
event_id	UUID v4	Globally unique event identifier.
event_type	string	The event type from this catalogue.
ustn	string	The USTN of the trade this event belongs to.
parent_event_id	UUID v4 or null	The event that this event is a response to (if applicable).
timestamp	ISO 8601 UTC	Event creation timestamp.
authority_class	enum	Observation, Assertion, or Confirmation.
authority_party_id	string	The party recording the event.
authority_signature	string	Cryptographic signature over the event content.
hash_prev	hex	SHA-256 hash of the previous event in the spine.
hash_self	hex	SHA-256 hash of this event (computed after all other fields are set).
governor_decision_ref	UUID v4	Reference to the Governor decision that authorised this event.
loom_entry_ref	UUID v4	Reference to the Loom audit log entry for this event.

Event Type Catalogue (selected; full set per source)

[L2] The table below lists representative event types from the catalogue. The full catalogue contains fifty or more event types; the source documents preserve the complete list. Each event type has a canonical name (EVT_*) and a defined set of additional fields beyond the common required fields.

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Event Type	Required Fields (additional)	Authority	Clock Impact	Lineage
EVT_TRADE_CREATE_D	buyer_party_id, seller_party_id, contract_ref	Assertion	Commercial: F1	Depends on: counterparty relationship exists.
EVT_CONTRACT_LOCKED	contract_hash, contract_terms_hash	Assertion	Commercial: F2	Depends on: EVT_TRADE_CREATED.
EVT_USTN_MINTED	ustn, parent_ustn (if multi-shipment)	Assertion	Governance: F2	Depends on: EVT_CONTRACT_LOCKED.
EVT_QUOTE_SUBMITTED	quote_id, quote_amount, quote_currency	Assertion	Commercial: F1	Depends on: EVT_TRADE_CREATED.
EVT_PAYMENT_LEG_PROPOSED	leg_id, payer, payee, amount, currency	Assertion	Financial: F1	Depends on: EVT_CONTRACT_LOCKED.
EVT_SETTLEMENT_INSTRUCTION_SIGNED	instruction_id, legs[], signature	Assertion	Financial: F2	Depends on: EVT_PAYMENT_LEG_PROPOSED.
EVT_SETTLEMENT_CONFIRMED	bank_tx_ref, executed_amount, executed_ts	Confirmation	Financial: F4	Depends on: EVT_SETTLEMENT_INSTRUCTION_SIGNED.
EVT_CUSTOMS_DECLARATION_FILED	entry_number, broker_licence	Assertion	Customs: F2	Depends on: EVT_CONTRACT_LOCKED.
EVT_CUSTOMS_CLEARANCE_GRANTED	clearance_ref, clearing_authority	Confirmation	Customs: F4	Depends on: EVT_CUSTOMS_DECLARATION_FILED.
EVT_PICKUP_CONFIRMED	carrier_id, vehicle_id, pickup_ts	Assertion	Logistics: F2	Depends on: EVT_USTN_MINTED.
EVT_IN_TRANSIT	mode, location, transit_ts	Observation	Logistics: F2	Depends on: EVT_PICKUP_CONFIRMED.
EVT_ARRIVAL_AT_PORT	port_id, arrival_ts	Observation	Logistics: F3	Depends on: EVT_IN_TRANSIT.
EVT_DELIVERY_ACCEPTED	consignee_id, delivery_ts, acceptance_signature	Confirmation	Logistics: F4	Depends on: EVT_ARRIVAL_AT_PORT.
EVT_DOCUMENT_ISSUED	doc_type, doc_id, issuer_id	Assertion	Documentation: F2	Depends on: EVT_CONTRACT_LOCKED.
EVT_DOCUMENT_VERIFIED	doc_id, verifier_id, verification_ts	Confirmation	Documentation: F3	Depends on: EVT_DOCUMENT_ISSUED.
EVT_INSPECTION_PERFORMED	inspection_id, agency_id, result	Assertion	QC: F2	Depends on: EVT_ARRIVAL_AT_PORT.
EVT_INSPECTION_CERTIFICATE_ISSUED	cert_id, agency_id, cert_hash	Confirmation	QC: F3	Depends on: EVT_INSPECTION_PERFORMED.
EVT_INSURANCE_POLICY_BOUND	policy_id, insurer_id, coverage_amount	Assertion	Insurance: F2	Depends on: EVT_CONTRACT_LOCKED.
EVT_INSURANCE_CLAIM_FILED	claim_id, claim_amount, claim_reason	Assertion	Insurance: F2	Depends on: EVT_INSURANCE_POLICY_BOUND.
EVT_COMPLIANCE_SCREENING_PASSED	screening_id, result, screened_parties[]	Assertion	Compliance: F2	Depends on: EVT_TRADE_CREATED.
EVT_COMPLIANCE_ALERT_RAISED	alert_id, alert_type, severity	Assertion	Compliance: F2	Depends on: EVT_COMPLIANCE_SCREENING_PASSED.

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Event Type	Required Fields (additional)	Authority	Clock Impact	Lineage
EVT_DISPUTE_OPENED	dispute_id, opener_party_id, dispute_subject	Assertion	Dispute: F2	Depends on: any post-contract event.
EVT_DISPUTE_RESOLVED	dispute_id, resolution_type, resolution_ref	Confirmation	Dispute: F4	Depends on: EVT_DISPUTE_OPENED.
EVT_EVIDENCE_PACKAGE_ASSEMBLED	package_id, evidence_refs[]	Assertion	Evidence: F2	Depends on: multiple prior events.
EVIDENCE_SEALED	package_id, seal_signature, seal_ts	Confirmation	Evidence: F5	Depends on: EVT_EVIDENCE_PACKAGE_ASSEMBLED.
EVT_CLOSURE_REQUESTED	closure_request_id, requested_by	Assertion	Governance: F1	Depends on: all 7 closure conditions per Part 8.
EVT_CLOSURE_GRANTED	closure_decision_ref, multisig_refs[]	Confirmation	Governance: F5	Depends on: EVT_CLOSURE_REQUESTED, canClose=true.
EVT_CLOSURE_WITH_EXCEPTION	exceptions[], severities[]	Confirmation	Governance: F5	Depends on: EVT_CLOSURE_REQUESTED, severity<=2.
EVT_GOVERNOR_DECISION	decision_id, gate_decisions[], merged_decision	Assertion	Governance: F3	Depends on: command being evaluated.
EVT_MULTISIG_OBTAINED	multisig_id, signers[], threshold	Confirmation	Governance: F4	Depends on: EVT_GOVERNOR_DECISION.
EVT_AMENDMENT_PROPOSED	amendment_id, proposed_text, affected_L0[]	Assertion	Governance: F1	Depends on: constitutional signatory status.
EVT_AMENDMENT_SIGNED	amendment_id, multisig_refs[], wasm_module_hash	Confirmation	Governance: F5	Depends on: EVT_AMENDMENT_PROPOSED + 30-day notice.
EVT_RECOVERY_INITIATED	recovery_id, failed_event_ref, recovery_strategy	Assertion	Governance: F2	Depends on: a failed or partial state.
EVT_RECOVERY_COMPLETED	recovery_id, recovered_state_hash	Confirmation	Governance: F3	Depends on: EVT_RECOVERY_INITIATED.
EVT_OBSERVATION_EVENT	observation_type, observed_data, source	Observation	varies	Depends on: relevant prior event.
EVT_POST_CLOSURE_SUPPLEMENT	sealed_package_ref, supplement_package_id	Assertion	Evidence: F2	Depends on: EVT_EVIDENCE_SEALED.
EVT_REGULATORY_INQUIRY	inquiry_id, regulator_id, inquiry_scope	Observation	Governance: F1	Depends on: trade in closed state.
EVT_REGULATORY_RESPONSE	inquiry_id, response_data, response_signature	Assertion	Governance: F2	Depends on: EVT_REGULATORY_INQUIRY.
EVT_DATA_RESIDENCY_CHECK	data_element_id, residency_class, storage_location	Observation	Compliance: F2	Depends on: data write.
EVT_AUDIT_ANNOTATION	annotated_event_ref, annotation_text, annotator_id	Observation	Governance: F1	Depends on: any prior event.

[L2] The catalogue is normative: an event type not in this catalogue may not be recorded on the spine. Adding a new event type is a governed change: it requires an update to the catalogue (itself a governed artefact), an implementation of the event type's validator, and a corresponding OPA policy. The change

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does not require constitutional amendment (the catalogue is L2, not L0), but it does require Governor approval.

Part 12 — Canonical Data Model Outline

[L2] The Canonical Data Model defines the persistent data structures of the SGTX platform. The model is outlined here; detailed field specifications are preserved in the source documents. The model is organised into seven major components, each addressing a specific persistence concern.

1. State Vector Store (12 domains x F0-F5)

[L2] The State Vector Store is the persistent representation of the multi-clock state vector [see Part 1]. For each trade, the store maintains the current finality class of each of the twelve domain clocks, the timestamp of the last finality advancement per domain, and the event references that drove each advancement. The store is a derived view: it is rebuilt from the event spine on replay [see Part 2, Replay Mode].

2. Event Spine (hash-chained, immutable)

[L2] The Event Spine is the persistent, append-only, hash-chained store of every event recorded in the platform. Events are represented per the Canonical Event Type Catalogue [see Part 11]. The spine is the authoritative history; all other stores are derived from it. The spine is replicated to a quorum of audit nodes; the quorum protocol is specified in Part 14.

3. Obligation Graph (directed dependency graph)

[L2] The Obligation Graph is the directed dependency graph of all obligations in the platform. An obligation is a commitment that one party owes to another (e.g., a payment obligation, a delivery obligation, a documentation obligation). The graph records the dependencies between obligations: which obligations must be discharged before which others. The graph is used by the Governor to evaluate closure readiness [see Part 8] and by the settlement orchestration to determine sequencing [see Part 4].

4. Settlement Instructions and Legs

[L2] The Settlement Instructions and Legs store is the persistent representation of settlement instructions and their constituent payment legs [see Part 4]. Each instruction and leg is recorded with its full provenance: originator, governor decision reference, executing bank, execution timestamp, and settlement confirmation.

5. External Identifier Registry (17 identifier types)

[L2] The External Identifier Registry is the persistent mapping between SGTX internal identifiers (USTNs, event IDs) and external identifiers (carrier booking numbers, customs entry numbers, bank transaction references, B/L numbers, AWB numbers, etc.). The registry preserves the authority of external identifiers per C-20 (USTN as Namespace, Not Override). Seventeen identifier types are currently registered; the registry is extensible.

#	Identifier Type	External Authority
1	Carrier Booking Number	Carrier
2	Bill of Lading (B/L) Number	Carrier
3	Air Waybill (AWB) Number	Air carrier
4	Railway Consignment Note Number	Rail carrier
5	RoRo B/L Number	RoRo carrier

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#	Identifier Type	External Authority
6	Customs Entry Number	Customs authority
7	Customs Clearance Reference	Customs authority
8	Bank Transaction Reference	Executing bank
9	LC Number	Issuing bank
10	Insurance Policy Number	Insurer
11	Inspection Certificate Number	Inspection agency
12	Port Terminal Receipt Number	Terminal operator
13	Container Number	Container owner
14	Vehicle Identification Number (VIN)	Vehicle manufacturer
15	Government Trade ID	Government authority
16	Tax Invoice Number	Tax authority
17	Quotation Reference	Issuing party

6. Recovery Vault (content-addressable)

[L2] The Recovery Vault is the content-addressable store of recovery materials. When a state-changing operation fails or partially completes, the recovery materials (the operation context, the intermediate state, the failure reason) are stored in the vault, addressed by the content hash of the materials. The vault enables recovery without erasure (per C-19): the failed path is preserved; the recovery path builds on it.

7. Transaction Twin (14 domains)

[L2] The Transaction Twin is the digital twin of a trade: a real-time, queryable representation of the trade's current state across fourteen domains (the twelve state vector clocks plus two cross-cutting domains: Risk and Audit). The twin is used by operators, auditors, and AI subsystems to query the current state of a trade without traversing the event spine. The twin is a derived view; it is rebuilt from the spine on demand or on material state change.

Part 13 — API Contract Structure

[L2] The API Contract Structure defines the external API surface of the SGTX platform. The API follows the command/event separation (per C-22): commands are submitted via the API; events are consumed via subscriptions or queried via read APIs. The API is RESTful for resource operations, with an additional event subscription channel for real-time event delivery.

Command / Event Separation in the API

Aspect	Command API	Event API
HTTP method	POST (mutates state)	GET (reads state) or SUBSCRIBE (stream)
Idempotency	Required (idempotency key)	N/A (events are immutable)
Response	202 Accepted with command_id	200 OK with event payload, or stream
Failure mode	400/422 with error detail	N/A (events cannot fail to read)
Authorisation	Per-command authorisation	Per-stream authorisation

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REST Outline for Major Resources

Resource	Methods	Description
/trades	POST, GET	Create a new trade (command); list trades (read).
/trades/{ustn}	GET	Get a single trade by USTN.
/trades/{ustn}/state	GET	Get the current state vector of a trade.
/trades/{ustn}/events	GET, SUBSCRIBE	List events for a trade; subscribe to new events.
/trades/{ustn}/closure	POST, GET	Request closure (command); check closure status (read).
/settlement-instructions	POST, GET	Submit a settlement instruction (command); list instructions.
/settlement-instructions/{id}	GET	Get a single instruction.
/payment-legs/{id}	GET	Get a single payment leg.
/obligations	GET	List obligations; filter by trade, party, or status.
/external-identifiers	GET, POST	Lookup by external ID; register a new mapping.
/governor/decisions/{id}	GET	Get a single Governor decision.
/audit/loom	GET	Query the Loom audit log (read-only).

Idempotency Middleware

[L2] All command APIs require an idempotency key in the request header. The idempotency key is a client-generated UUID; the server stores the response for the key for a configurable period (default: 24 hours). If the same key is reused within the period, the server returns the stored response without re-executing the command. This protects against network retries causing duplicate state changes.

Rate Limiting

[L2] All APIs are rate-limited per client identity. The default limits are: 100 commands per minute per tenant; 1000 reads per minute per tenant; 10 event subscriptions per tenant. Limits are configurable per tenant and per API. Rate-limit exceeded responses use HTTP 429 with a Retry-After header.

Part 14 — Observability Catalogue

[L2] The Observability Catalogue defines the metrics, failure budgets, and failure-path test suite for the SGTX platform. Observability is a first-class concern: the platform is designed to be auditable in real time, not just post-hoc.

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Metrics

Metric	Type	Description
sgtx_trade_count_total	counter	Total number of trades created.
sgtx_trade_closure_rate	gauge	Fraction of trades that reached CLOSED state.
sgtx_governor_decision_latency_seconds	histogram	Latency of Governor decisions (intake to commit).
sgtx_event_spine_append_latency_seconds	histogram	Latency of event spine appends.
sgtx_loom_replication_lag_seconds	gauge	Lag between primary and quorum Loom replicas.
sgtx_settlement_instruction_count_total	counter	Total settlement instructions submitted.
sgtx_settlement_confirmation_latency_seconds	histogram	Latency from instruction submission to bank confirmation.
sgtx_closure_with_exception_count_total	counter	Total closures with exceptions, by severity.
sgtx_divergence_index_distribution	histogram	Distribution of divergence indices across active trades.
sgtx_ai_recommendation_count_total	counter	Total AI recommendations, by acceptance/rejection.
sgtx_wasmedge_veto_count_total	counter	Total WasmEdge vetoes (constitutional violations caught).
sgtx_opa_policy_evaluation_seconds	histogram	Latency of OPA policy evaluations.
sgtx_quorum_node_count	gauge	Number of Loom quorum nodes currently healthy.
sgtx_recovery_initiated_count_total	counter	Total recovery initiations.
sgtx_recovery_completion_seconds	histogram	Latency of recovery completions.

Failure Budgets

Component	RTO	RPO	Failure Budget
Event Spine (primary)	4 hours	0 (append-only)	No data loss permitted; quorum replication ensures RPO=0.
Loom Audit Log (quorum)	4 hours	0	No data loss permitted; append-only with quorum replication.
State Vector Store	4 hours	<= 5 minutes	Rebuildable from event spine; small RPO acceptable.
Settlement Orchestration	4 hours	<= 1 minute	In-flight instructions must be recoverable.
Governor Pipeline	4 hours	<= 1 minute	In-flight commands must be recoverable.
API Gateway	1 hour	N/A (stateless)	Stateless; reroute to healthy instances.
AI Subsystem	24 hours	<= 1 hour	AI is advisory; degraded mode without AI is acceptable.

Minimum Failure-Path Test Suite

[L2] The minimum failure-path test suite is derived from the adversarial testing list. Each test exercises a specific failure path and verifies that the platform responds correctly (recovery without erasure, audit trail preserved, no state corruption). The suite is run continuously; failures block deployment.

Layer 2 | Implementation Specifications

- **Bank settlement rejection:** A bank rejects a settlement instruction; the platform records the rejection, marks the leg as failed, and triggers reconciliation.
- **Customs hold:** Customs places a hold on a declaration; the platform records the hold, suspends downstream obligations, and notifies the consignee.
- **Carrier no-show:** A carrier fails to pick up; the platform records the no-show, triggers rebooking, and adjusts the logistics clock.
- **Governor DENY:** The Governor denies a command; the platform records the denial with the denying gate and reason, and no state change occurs.
- **WasmEdge veto:** WasmEdge vetoes an OPA decision; the platform records the veto, blocks the operation, and raises a SEV-2 incident.
- **Loom quorum loss:** A quorum of Loom nodes becomes unavailable; the platform continues to operate in degraded mode (reads only) and blocks state-changing operations until quorum is restored.
- **AI subsystem failure:** The AI subsystem becomes unavailable; the platform continues to operate in deterministic-only mode (AI advisories are absent but Governor decisions proceed).
- **Partial settlement execution:** A multi-leg settlement instruction partially executes (some legs settle, some fail); the platform applies the atomicity policy and records the outcomes.
- **Closure with exception:** A trade is closed with exceptions; the platform verifies the severity matrix and records the closure with the exceptions and mitigations.
- **Concurrent Governor commands:** Two commands target the same trade concurrently; the platform serialises them via the event spine and records the order.

Part 15 — RTO/RPO Targets

[L2] The Recovery Time Objective (RTO) and Recovery Point Objective (RPO) targets define the platform's resilience commitments. Critical components have a 4-hour RTO; standard components have a 24-hour RTO. The event spine is append-only, so its RPO is 0 (no data loss is acceptable). Other components have small but non-zero RPOs reflecting their rebuildable nature.

RTO Targets

Tier	RTO	Components
Critical	4 hours	Event Spine, Loom Audit Log, Governor Pipeline, Settlement Orchestration, State Vector Store.
Standard	24 hours	AI Subsystem, Reporting, Analytics, Non-critical add-ons.
Stateless	1 hour	API Gateway, Web Frontend.

RPO Targets

Tier	RPO	Components
Zero-loss	0	Event Spine (append-only, hash-chained, quorum-replicated); Loom Audit Log (append-only, quorum-replicated).
Near-zero	<= 1 minute	Governor Pipeline (in-flight commands); Settlement Orchestration (in-flight instructions).
Small	<= 5 minutes	State Vector Store (rebuildable from event spine).
Moderate	<= 1 hour	AI Subsystem state; reporting caches.

Layer 2 | Implementation Specifications

Durability and Quorum Definitions

[L2] Durability is the guarantee that committed data is not lost. For the event spine and Loom audit log, durability is achieved through quorum replication: data is replicated to N nodes, and a write is acknowledged only after W nodes confirm (write quorum); a read is satisfied by R nodes (read quorum). The classic quorum formula is $W + R > N$, ensuring reads and writes overlap.

Parameter	Default	Meaning
N (replication factor)	5	Each event is replicated to 5 quorum nodes.
W (write quorum)	3	A write is acknowledged after 3 of 5 nodes confirm.
R (read quorum)	2	A read is satisfied by 2 of 5 nodes.
$W + R > N$	$3 + 2 = 5 > 5?$	$5 > 5$ is false; adjust R to 3 for strict quorum.

[L2] Note: with $N=5$, $W=3$, $R=3$, we have $W + R = 6 > N = 5$, satisfying strict quorum. The default configuration uses $W=3$, $R=3$ for the event spine and Loom audit log. Other components may use weaker quorum ($W=2$, $R=2$ for $N=5$) where reconstruction is possible.

APPENDICES

Appendices A through F

Supporting detail: financial control, regulatory, operating model, implementation priority, audit trail, and source manifest.

Appendix A — Financial Control Framework (CFO)

[L2] The Financial Control Framework defines the financial discipline of the SGTx platform. It covers the fee schedule, the reserve policy, the non-custody attestation template, unit economics, and capital and liquidity notes. The framework is the CFO's governing instrument; deviations require CFO approval and, where constitutional, constitutional amendment.

Fee Schedule

[L2] SGTx charges a fee on each trade. The fee is calculated as a percentage of the trade value, with a minimum fee per trade. The default fee schedule is:

Tier	Trade Value (USD)	Fee Rate	Minimum Fee (USD)
Micro	0 - 10,000	1.5%	50
Small	10,001 - 100,000	1.5%	150
Medium	100,001 - 1,000,000	1.5%	1,500
Large	1,000,001 - 10,000,000	1.2%	15,000
Mega	10,000,001+	1.0%	120,000

[L2] The fee is charged to the originating party at trade closure. Fees are invoiced monthly; payment terms are net-30. Fee schedules are configurable per tenant and per jurisdiction; deviations from the default require CFO approval.

Reserve Policy (110% Backing)

[L2] Per C-17 (110% Reserve Rule), every USTN-backed reserve instrument is backed at no less than 110%. The reserve composition is governed by the CFO; the composition rules are:

- **Cash component ($\geq 80\%$):** Cash held at licensed banks in the instrument's denominated currency. Must be available for settlement within 24 hours.
- **Government securities ($\leq 20\%$):** Short-duration government securities (T-bills or equivalent) rated AA- or higher. Used for yield; not for speculative purposes.
- **Forbidden assets:** Equity, corporate bonds below AA-, cryptocurrencies, commodities, derivatives (other than for hedging the reserve itself).

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Reserve Composition Rules

Asset Class	Minimum	Maximum	Notes
Cash at licensed banks	80%	100%	Same currency as the instrument; same jurisdiction where possible.
Government securities (AA- or higher)	0%	20%	Short duration (≤ 90 days); same currency as the instrument.
Other	0%	0%	Forbidden.
Total	100%	110%+	The 10% buffer above 100% provides resilience against valuation fluctuations.

ZK Attestation

[L2] The reserve ratio is attested cryptographically using zero-knowledge proofs (Add-On 7). The proof demonstrates that the reserve ratio is at least 110% without revealing the exact composition or the exact ratio. The proof is published on a regular cadence (default: daily); the proof verification key is public; the underlying reserve data is attested but not disclosed in detail. This provides transparency on the ratio while preserving operational confidentiality on the composition.

Non-Custody Attestation Template

NON-CUSTODY ATTESTATION

Date: <YYYY-MM-DD>

Attesting Party: <SGTX operator entity>

Attestation Period: <start> to <end>

I attest that, for the attestation period:

1. SGTX did not take custody of customer funds at any point.
2. All fiat settlement was executed by licensed banks on each payment leg; SGTX orchestrated instructions only.
3. SGTX did not maintain any internal account capable of holding customer funds.
4. All reserve instruments were held at licensed institutions; SGTX did not itself hold reserve assets.
5. The reserve ratio for each USTN-backed reserve instrument was at least 110% throughout the attestation period, as demonstrated by the published ZK attestation proofs.

Signature: <signatory>

Witness (independent auditor): <auditor>

Unit Economics Illustration

[L2] The unit economics of a single trade are illustrated below. The example uses a Medium-tier trade (USD 500,000 trade value) with default fee schedule and standard operational costs. The illustration is for clarity; actual economics vary per tenant, per jurisdiction, and per trade complexity.

Item	Amount (USD)	Notes
Trade value	500,000.00	Medium tier.
SGTX fee (1.5%)	7,500.00	Charged to originating party at closure.
Bank settlement fee	250.00	Paid to executing bank(s); passes through.
Customs broker fee	500.00	Paid to broker; passes through.

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Item	Amount (USD)	Notes
Carrier fee	8,000.00	Paid to carrier; passes through.
SGTX gross revenue	7,500.00	Fee revenue.
SGTX operational cost (allocated)	1,200.00	Infrastructure, AI inference, support allocation.
SGTX net contribution	6,300.00	Before reserve buffer allocation.
Reserve buffer allocation (10% of fee)	750.00	Held in reserve to maintain 110% backing.
SGTX net after reserve	5,550.00	Net contribution to platform.

Capital and Liquidity Notes

[L2] SGTX is non-custodial and does not hold customer funds. However, SGTX does maintain operational capital for: (i) reserve buffer allocations (per the 110% rule); (ii) operational contingency (to cover operational failures and reconcilements); and (iii) regulatory capital where required by jurisdiction. The capital structure is governed by the CFO; the liquidity policy ensures that operational capital is available within 24 hours and reserve buffers are available within 7 days.

Appendix B — Regulatory and Legal

[L2] This appendix provides the regulatory classification matrix, the data-residency element classification, and the SAR/FIU (Suspicious Activity Report / Financial Intelligence Unit) interface notes. The materials here are operational references; they do not constitute legal advice. Final regulatory determinations are made by the relevant regulators per the Regulatory Classification Gate [see Part 7].

Regulatory Classification Matrix (Egypt / CBE Law 194/2020 initial)

[L2] The initial reference jurisdiction is Egypt, with the Central Bank of Egypt (CBE) Law 194/2020 as the initial regulatory reference. The matrix below records the regulatory classification of each SGTX capability in Egypt. The matrix is extensible: as SGTX enters additional jurisdictions, their classifications are added to the same matrix.

Capability	Egypt Classification	Licence / Authorisation
Settlement orchestration	Payment service provider (per CBE Law 194/2020)	CBE authorisation; partnership with licensed Egyptian bank.
Customs declaration filing	Customs broker	Egyptian customs broker licence.
Cargo insurance binding	Insurance intermediary	Egyptian Financial Supervisory Authority (EFSA) intermediary licence.
Trade finance documentation	Banking service	Partnership with licensed Egyptian bank; no standalone SGTX licence.
Data processing	Data controller / processor	Egyptian Data Protection Law registration; data localisation per Law 151/2020.
GNN risk engine	Internal risk tool (not separately regulated)	No standalone licence; subject to bank partner's risk management oversight.

Data-Residency Element Classification

[L2] Data elements in SGTX are classified by residency class per C-28 (Data-Residency Respecting). The classification determines where the element may be stored and to where it may be replicated. Five residency classes are defined:

Class	Meaning	Example Elements
EGYPT_ONLY	Element may be stored only in Egypt. Replication outside Egypt is forbidden. Applies to Egyptian personal data and certain regulated trade data per Egyptian law.	Egyptian personal data; Egyptian customs declaration data.
COUNTRY_ONLY	Element may be stored only in the relevant country (per the data element's jurisdiction). Generalisation of EGYPT_ONLY.	Country-specific personal data; country-specific regulatory filings.
REGIONAL	Element may be stored and replicated within a defined region (e.g., MENA, EU). Replication outside the region is forbidden.	Regional trade data; regional regulatory data.
APPROVED_CROSS_BORDER	Element may be transferred cross-border to approved jurisdictions with adequate data protection. Approval is per the relevant jurisdiction's data export rules.	Cross-border trade documents; cross-border payment references.
GLOBAL_ALLOWED	Element may be stored and replicated globally. No residency restriction.	Anonymised aggregate metrics; public reference data.

Finding A-14 — Egypt Data Localisation vs Multi-Region Mirroring

[L2] The v13.0 audit clarified that the Main Blueprint's reference to *multi-region mirroring* does not override Egyptian data localisation requirements under Law 151/2020. Egyptian-resident data elements classified EGYPT_ONLY may not be mirrored to other regions, even for resilience. Resilience for EGYPT_ONLY elements is achieved through in-region replication (multiple Egyptian availability zones) rather than cross-region mirroring.

SAR / FIU Interface Notes

[L2] Where SGTX detects suspicious activity (per the compliance screening, Add-On 1 GNN Risk Engine and Add-On 18 Trade Compliance Calendar), it must be able to file a Suspicious Activity Report (SAR) with the relevant Financial Intelligence Unit (FIU). The SAR/FIU interface is jurisdiction-specific; the interface notes below describe the general pattern, with Egypt-specific examples.

- **Detection:** The compliance screening raises an alert; the alert is recorded as EVT_COMPLIANCE_ALERT_RAISED on the event spine.
- **Investigation:** A compliance officer reviews the alert. The investigation is recorded as a governed artefact.
- **SAR filing decision:** If the investigation concludes that a SAR is warranted, the compliance officer files the SAR with the relevant FIU (in Egypt: the Egyptian Money Laundering Combating Unit, EMLCU).
- **SAR record:** The SAR filing is recorded as an event on the event spine; the SAR document itself is stored in the Evidence Store with restricted access.
- **Confidentiality:** SAR filings are confidential; access is restricted to authorised compliance personnel and regulators. The event spine records the SAR existence but not the SAR content (which is in the restricted Evidence Store).

Appendix C — Operating Model

[L2] The Operating Model defines the governance, multisig, change-management, and incident severity arrangements for the SGTX platform. It is the operational counterpart to the constitutional and architectural layers.

Platform Governance Authority RACI

[L2] The Platform Governance Authority (PGA) is the cross-functional body responsible for the SGTX platform. The RACI matrix below identifies the Responsible, Accountable, Consulted, and Informed parties for each major platform activity.

Activity	PGA Chair	CTO	CFO	CCO	CISO	Legal
Constitutional amendment	A	C	C	C	C	R
Architectural change (L1)	A	R	C	C	C	I
Implementation change (L2)	I	A/R	I	I	C	I
OPA policy update	I	A	C	R	C	I
WasmEdge module update	A	R	I	I	C	C
Add-on activation	A	R	C	C	C	C
Jurisdiction expansion	A	C	C	R	C	R
Incident response (SEV-0/1)	A	R	C	C	R	C
Incident response (SEV-2+)	I	A	C	C	R	I
Multisig authorisation	A	R	C	C	C	C

R = Responsible, A = Accountable, C = Consulted, I = Informed

Multisig Policy

[L2] Per C-25 (Multisig for Irreversible), operations classified as irreversible require multisig authorisation. The multisig thresholds are:

Operation Class	Threshold	Signatory Pool
Standard operations (settlement release, USTN sealing, evidence destruction)	2-of-3	Three designated operations signatories (e.g., CTO, COO, CFO or their delegates).
Constitutional amendments	3-of-5	Five designated constitutional signatories (the above three plus Chief Legal Officer and Chief Compliance Officer).
Emergency operations (incident response)	2-of-3	Three designated incident response signatories (CTO, CISO, COO or delegates).

Change-Management Process for WASM Modules and Policies

[L2] Changes to WasmEdge modules (L0 constitutional encodings) and OPA policies (L1 architectural encodings) follow a governed change-management process. The process ensures that changes are reviewed, signed, versioned, and auditable.

- 1 Proposal:** A change proposal is drafted, identifying the affected module/policy, the change text, the rationale, and the impact assessment.

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- 2 Review:** The proposal is reviewed by the PGA. For WasmEdge changes, the review includes a constitutional review (does the change amend an L0 statement? If so, the constitutional amendment process applies).
- 3 Approval:** The proposal is approved by the relevant authority (PGA for L1 changes; constitutional signatories for L0 changes).
- 4 Implementation:** The change is implemented in the module or policy. The implementation is reviewed and signed.
- 5 Deployment:** The signed module or policy is deployed. The previous version is archived; the new version is loaded; the loading event is recorded in the Loom audit log.
- 6 Verification:** Post-deployment verification confirms that the new module or policy is operating as expected. Verification failures trigger rollback (for L2 changes) or emergency review (for L0/L1 changes).

Incident Severity Model

Severity	Definition	Response	Escalation
SEV-0	Critical: platform-wide outage; data loss; constitutional violation in production.	Immediate; incident response team activated; war room.	CEO, CTO, CISO, CCO, Legal; constitutional signatories for constitutional violations.
SEV-1	High: major functionality unavailable; settlement failures; multiple trades affected.	Within 1 hour; incident response team engaged.	CTO, CISO, COO; PGA chair for prolonged incidents.
SEV-2	Medium: significant functionality degraded; isolated settlement failure; single trade affected materially.	Within 4 hours; on-call engineer engaged.	CTO, CISO; PGA chair notified.
SEV-3	Low: minor functionality degraded; workaround available.	Within 24 hours; routed to engineering queue.	Engineering manager.
SEV-4	Cosmetic: UI issues; documentation errors; non-functional defects.	Within 7 days; routed to product backlog.	Product manager.

Appendix D — Implementation Priority Framework

[L2] The Implementation Priority Framework defines the order in which capabilities are implemented. The framework is dependency-driven: capabilities are prioritised based on their dependencies, not on calendar dates. No calendar claims are made; implementation proceeds as dependencies are satisfied.

Priority Waves

Wave	Scope	Dependencies
P0	Core trade execution (Trade, Contract, USTN, Governor, Event Spine).	Foundation: must be implemented first; everything else depends on it.
P1	Transport engines (Road, Air, Ocean Container, RoRo, Rail).	Depends on P0 (USTN, Governor, Event Spine).
P2	Add-Ons 1 through 8 (critical intelligence + compliance).	Depends on P0 (Governor, Event Spine); some add-ons depend on P1 transport engines.
P3	Add-Ons 9 through 18 (operational add-ons).	Depends on P0 and P1; some add-ons depend on P2 (e.g., demurrage depends on settlement orchestration).
P4	Add-Ons 19 through 28 (extended functionality + GRiRE).	Depends on P0, P1, P2, P3 as applicable; GRiRE (Add-On 28) depends on the jurisdiction adapter schema (Part 6).

P0 — Core Trade Execution

[L2] P0 implements the foundational capabilities: the Trade entity, the Contract lock, the USTN minter, the Governor pipeline (all seven gates), and the Event Spine. P0 is the floor: nothing else functions without P0. P0 status: CORE_READY (internal architecture specified and verified).

P1 — Transport Engines

[L2] P1 implements the five transport engines (Road, Air, Ocean Container, RoRo, Rail) and the Multimodal Orchestrator. Each engine implements the mode-specific data model [see Part 10] and integrates with the relevant external carrier systems. P1 status: mixed; some engines CORE_READY, others in active development.

P2 — Add-Ons 1-8 (Critical Intelligence + Compliance)

[L2] P2 implements the eight critical intelligence and compliance add-ons: GNN Risk Engine, Federated Learning, Causal Inference Engine, Self-Healing Infrastructure, Automated Penetration Testing, Post-Quantum Cryptography, Expanded ZK Proofs, and Customs Bond and Guarantee Management. These add-ons are foundational for risk management, security, and compliance.

P3 — Add-Ons 9-18 (Operational Add-Ons)

[L2] P3 implements the ten operational add-ons: Demurrage and Detention, Broker Liability and Insurance, Customs Valuation Intelligence, Cold Chain Quality Management, Inspection Agency Accreditation, Currency Risk Management, Government API Sandbox, FTA Preference Management, Piracy and Security Risk Engine, and Trade Compliance Calendar. These add-ons extend the platform with operational capabilities required for production trade execution.

P4 — Add-Ons 19-28 (Extended Functionality + GRiRE)

[L2] P4 implements the ten extended-functionality add-ons: Cargo Insurance Integration, Trade Finance Documentation, Back-to-Back LC Management, Force Majeure Handling, Shipper's Declaration and Export Docs, Port and Terminal Integration, Payment Guarantee Confirmation, Demurrage Dispute Resolution, the reserved slot (Add-On 27), and GRiRE (Global Regulatory Intelligence and Requirements Engine). GRiRE is the capstone: it depends on the jurisdiction adapter schema (Part 6) and provides the regulatory intelligence that keeps the platform current as jurisdictions evolve their regulations.

Appendix E — Full Historical Audit Trail (Immutable Annex)

[L2] This appendix preserves the full v13.1 FINAL audit trail verbatim. Nothing is removed; nothing is rewritten retrospectively. The v13.1 Part A audit (findings A-01 through A-24) is preserved in summary form below; the v12.0 audit header is preserved thereafter; the change log from v13.1 to v14.0 closes the appendix and is also captured in Section 8 of this document.

Immutability of the Audit Trail

[L2] The audit trail is an immutable annex. The findings, resolutions, and integration impacts recorded here are the historical record; they may not be modified except through explicit supersession in a later edition, and any supersession is itself recorded. This is the architectural realisation of C-19 (Recovery Is Not Erasure) applied to the document itself.

E.1 — v13.1 Part A Audit (A-01 through A-24, Summary)

[L2] The v13.0 re-audit identified twenty-four findings. Each finding is summarised below with its conflict description, canonical resolution, and integration impact. The full text of each finding is preserved in the source documents (Source 1 and Source 2; see Appendix F).

#	Title	Conflict	Resolution	Integration Impact
A-01	Direct Government API Scope	The Main Blueprint states that SGTx directly connects to only four government APIs (Nafeza, CargoX, ETA, CBE). The change-set establishes a worldwide country-adaptor and multi-agency gateway architecture that is materially broader than the four-connector reference set.	Retain the four direct Egypt integrations as the initial / reference direct-integration set. Add the worldwide adaptor / gateway fabric as the extensibility layer. The term "Direct API" therefore denotes a currently adopted first-party connector, not a limitation on the global adaptor model.	The Main Blueprint body retains all original Nafeza, CargoX, ETA and CBE material unchanged. The change-set material on the worldwide country-adaptor and gateway architecture is preserved verbatim. Add-On 15 (Government API Sandbox) and Add-On 28 (GRiRE) provide the detailed specification for the extensibility layer.
A-02	AI A4 Authority Language	The Main Blueprint uses the phrase "A4 Governance / auto-executes within constitutional bounds," while the change-set explicitly states that AI never has independent authoritative decision rights.	Clarify that A4 is deterministic policy automation executed by the Governor component together with OPA and WasmEdge under pre-authorised rules. The AI subsystem itself never acquires independent execution authority; it proposes, explains, classifies, and escalates, while the deterministic policy engine and authorised institutions execute. This preserves the A4 low-risk automation capability while making the constitutional boundary explicit and auditable.	The A4 rung of the AI Authority Ladder is clarified in L0 (A4: Execution Within Bounds). The architectural realisation is in L1, Part 9 (AI Recommendation Gateway). The A4 clarification is now governing language.

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#	Title	Conflict	Resolution	Integration Impact
A-03	Non-Custody vs Payment / Settlement Functionality	The Main Blueprint describes non-custody as a defining property while also describing payment/settlement functionality. The change-set seeks to clarify that non-custody is compatible with payment/settlement functionality provided the payment/settlement is executed by licensed institutions.	Non-custody is an architectural property (no SGTX-internal account can hold customer funds), not a legal classification. Non-custody is compatible with payment/settlement functionality provided the payment/settlement is executed by licensed institutions and SGTX only orchestrates instructions. Non-custody is not the absence of payment functionality; it is the absence of SGTX-internal fund holding.	The Non-Custody Principle is enshrined at L0 (C-01). The architectural realisation is in L1, Part 4 (Settlement Orchestration Control Plane). The settlement orchestration is non-custodial by design.
A-04	Stablecoin / DeFi References vs Bank-Authoritative Settlement	The Main Blueprint contains references to stablecoin / DeFi concepts that the change-set explicitly supersedes with bank-authoritative settlement.	SGTX does not issue stablecoins (C-30) and does not integrate with DeFi protocols (C-31). References to stablecoin-like instruments in earlier documents refer to USTN-backed reserve instruments held at licensed banks, not to SGTX-issued cryptographic tokens on a public blockchain. Settlement is bank-authoritative (G7, C-21).	The L0 constitution explicitly forbids stablecoin issuance and DeFi integration. The settlement architecture (L1, Part 4) is bank-authoritative throughout. Earlier references to stablecoins are preserved in the audit trail for traceability but are superseded by the canonical interpretation.
A-05	Reserve Metadata vs Custody	The Main Blueprint refers to reserve metadata in a way that could be misread as SGTX custody of reserves. The change-set clarifies that reserves are held at licensed institutions, not by SGTX.	Reserve instruments are held at licensed institutions (banks, custodians). SGTX maintains metadata about the reserves (composition, ratio, location) and publishes zero-knowledge attestations (Add-On 7), but does not itself hold reserve assets. The reserve metadata is data about reserves, not the reserves themselves.	The reserve policy (Appendix A) specifies that reserves are held at licensed institutions. The ZK attestation (Add-On 7) demonstrates the reserve ratio without SGTX taking custody of the underlying assets.

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#	Title	Conflict	Resolution	Integration Impact
A-06	Reserve Ratio Thresholds	The Main Blueprint establishes a 110% reserve rule. The change-set operationalises the rule with attestation and ZK evidence. There is no conflict; the change-set provides the evidence mechanism for the constitutional threshold.	No conflict requiring replacement. The change-set operationalises the existing 110% rule with attestation and ZK evidence. The pre-existing constitutional rule remains in force unless separately amended by the constitution. The two formulations are read as the same rule expressed at two layers: the constitutional layer sets the threshold; the change-set layer provides the cryptographic evidence that the threshold is met.	Both the constitutional 110% rule (C-17) and the ZK reserve attestation flow (Add-On 7, Appendix A) are preserved unchanged. Add-On 7 is interpreted as the evidence mechanism for the constitutional threshold.
A-07	GNN / Institutional Graph and Non-Marketplace Safeguards	The Main Blueprint describes a GNN-based institutional graph. The change-set ensures that the graph operates only on relationship-controlled data and does not introduce marketplace dynamics.	The GNN Risk Engine (Add-On 1) operates on the relationship-controlled trade graph. It does not introduce anonymous matching, does not rank counterparties for public selection, and does not facilitate trade introduction. The graph is built from established counterparty relationships; it is not a marketplace graph. All add-ons are non-marketplace bounded (C-27).	Add-On 1 is preserved in full in L1, Part 5. The non-marketplace boundary is enshrined at L0 (C-27) and is reflected in the add-on status matrix (NMB column: true for all twenty-eight add-ons).
A-08	"Seven Critical Add-Ons" vs Expanded Add-On Set	The Main Blueprint refers to "seven critical add-ons" in some passages. The change-set expands the catalogue to twenty-eight add-ons. The conflict is terminological: are the original seven replaced, retained, or supplemented?	Retain the original seven as the historically and constitutionally referenced set. Expand the add-on catalogue rather than renaming or deleting them; Add-On 8 and all subsequent modules (through Add-On 28) become additional canonical extensions. The historical "seven critical add-ons" wording is retained as a compatibility reference; the canonical catalogue is the full set of twenty-eight add-ons defined in the change-set.	The Main Blueprint references to "seven critical add-ons" are preserved. The complete twenty-eight-add-on catalogue is now present in full in L1, Part 5. As part of the v13.0 re-integration (see finding A-24), Add-Ons 9 through 28, which were absent from the v12.0 appended amendment body, have been inserted in their canonical position between Add-On 8 and the Final Summary.
A-09	RoRo as Ocean Submode vs First-Class Mode	The Main Blueprint in some passages treats RoRo as a submode of Ocean. The change-set treats RoRo as a first-class transport mode.	RoRo is a first-class transport mode, not a submode of Ocean Container. While both involve ocean shipping, the operational characteristics, documentation, and port infrastructure are distinct. v14.0 reflects this as five first-class modes, with RoRo elevated to its own mode.	The Transport Engine Architecture (L1, Part 10) lists five first-class modes: Road, Air, Ocean Container, RoRo, Rail. RoRo has its own mode-specific data model and its own integration surface.

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#	Title	Conflict	Resolution	Integration Impact
A-10	Egypt Nafeza as One Generic Integration vs Mode-Specific Applicability	The Main Blueprint describes Nafeza as a single integration. The change-set clarifies that Nafeza integration applicability is mode-specific (e.g., ocean container vs. air cargo have different Nafeza touchpoints).	The Nafeza integration is a single connector surface, but its applicability is mode-specific. The connector handles mode-specific message types and mode-specific customs workflows. The integration is one connector with multiple mode-specific adapters internally.	The Nafeza integration is preserved as a single connector in the Jurisdiction Capability Adapter Schema (L1, Part 6). Mode-specific applicability is captured in the connector's mode-specific adapter specifications.
A-11	Closure Semantics	The Main Blueprint describes closure in terms of time elapsed and certain conditions. The change-set clarifies that closure is earned only by completion of all closure conditions, not by time elapsed.	Closure is earned, not granted by time (C-18). A trade is CLOSED only when all seven closure conditions are met (L1, Part 8). Time elapsed does not grant closure. The canClose predicate is the deterministic evaluation of the seven conditions. CLOSED_WITH_EXCEPTION is permitted only for severity 1 and 2 exceptions.	The Closure Policy (L1, Part 8) specifies the seven closure conditions, the canClose predicate, and the CLOSED_WITH_EXCEPTION severity matrix. The closure semantics are now governing language.
A-12	USTN as Universal Reference vs External Authority	The Main Blueprint describes the USTN as a universal reference. The change-set clarifies that the USTN is a universal namespace within SGTX, not a supra-jurisdictional instrument that overrides external authority.	The USTN is a universal namespace within SGTX (C-14). It does not override external identifiers (carrier booking numbers, customs entry numbers, bank transaction references); those identifiers are preserved in the External Identifier Registry (L2, Part 12). The USTN is a namespace, not an override of external authority (C-20). A jurisdiction's customs authority, banking authority, or other external authority remains authoritative within its domain.	The USTN discipline is enshrined at L0 (C-14, C-15, C-20). The External Identifier Registry (L2, Part 12) preserves external identifiers alongside USTNs. The USTN-as-namespace interpretation is now governing language.

Appendices | Appendices A through F

#	Title	Conflict	Resolution	Integration Impact
A-13	ISO 2022 as Integration Output vs Canonical Data	The Main Blueprint describes ISO 2022 as a canonical data format in some passages. The change-set treats ISO 2022 as an integration output format (a translation target for bank connectors), not as the canonical internal data format.	ISO 2022 is an integration output format. SGTX's canonical internal data model is the Canonical Data Model (L2, Part 12). When SGTX sends a settlement instruction to a bank that requires ISO 2022, the instruction is translated to ISO 2022 at the connector boundary. The internal representation is not ISO 2022. This separation preserves SGTX's ability to integrate with banks that use other formats (e.g., SWIFT MT, proprietary APIs).	The Canonical Data Model (L2, Part 12) is the internal data model. ISO 2022 is referenced in the Bank Settlement Gateway (L1, Part 4) as a translation target, not as a canonical format. The integration-output interpretation is now governing language.
A-14	Egypt Data Localisation vs "Multi-Region Mirroring"	The Main Blueprint refers to multi-region mirroring for resilience. The change-set clarifies that Egyptian data localisation under Law 151/2020 forbids cross-region mirroring of Egyptian-resident data.	Multi-region mirroring does not override Egyptian data localisation. Egyptian-resident data elements (classified EGYPT_ONLY) may not be mirrored to other regions, even for resilience. Resilience for EGYPT_ONLY elements is achieved through in-region replication (multiple Egyptian availability zones) rather than cross-region mirroring. Data elements are classified by residency class (L2, Appendix B); the residency class governs the replication topology.	The data-residency element classification (L2, Appendix B) is preserved with five classes. The in-region replication requirement for EGYPT_ONLY elements is now governing language. The Main Blueprint's multi-region mirroring references are preserved for traceability but are superseded by the residency classification.
A-15	"Zero-Cost" Technology vs Real Institutional Costs	The Main Blueprint in some passages refers to "zero-cost" technology. The change-set clarifies that all integrations carry institutional cost and that the "zero-cost" framing is misleading.	The "zero-cost" framing is eliminated. All integrations carry institutional cost: bank fees, customs fees, carrier fees, inspection fees, insurance premiums, regulatory licensing costs, and operational costs. SGTX is transparent about these costs; the unit economics illustration (L2, Appendix A) shows the cost breakdown for a sample trade. The "institutional-cost scope clarified" framing replaces "zero-cost".	The "zero-cost" phrase is forbidden in v14.0 (see Section 3, Over-Claim Elimination). The unit economics illustration (L2, Appendix A) shows the actual cost structure. The institutional-cost framing is now governing language.

Appendices | Appendices A through F

#	Title	Conflict	Resolution	Integration Impact
A-16	Marketplace Terminology	The Main Blueprint occasionally uses the term "marketplace" in a colloquial sense (a place where commerce occurs). The change-set establishes that SGTX is not a marketplace in the narrow institutional sense (a public matching engine).	The colloquial usage of "marketplace" is eliminated. SGTX is a trade execution infrastructure, not a marketplace. The term "marketplace" is reserved for its narrow institutional meaning: a public matching engine. SGTX does not operate one (C-02). Trade relationships are relationship-controlled (C-16).	The Non-Marketplace Principle is enshrined at L0 (C-02). All twenty-eight add-ons are non-marketplace bounded (C-27; L1, Part 5, NMB column). The marketplace terminology is now governing language.
A-17	AI-Generated Explanations vs Authoritative Decisions	The Main Blueprint in some passages blurs the line between AI-generated explanations and authoritative decisions. The change-set establishes that AI-generated explanations are advisory (A1), not authoritative.	AI-generated explanations are advisory (A1). They may inform human decision-makers but do not bind them. Authoritative decisions are made by the Governor (deterministic OPA + WasmEdge) and by human decision-makers in their respective authority domains. The AI subsystem may explain its recommendations; the explanations are inputs to decision-making, not decisions themselves (C-09, C-26).	The AI Authority Ladder (L0, A0 through A5) and the AI Recommendation Gateway (L1, Part 9) are the architectural realisations. AI-generated explanations are clearly advisory in all v14.0 references.
A-18	Manual Fallback vs Automated Governance	The Main Blueprint in some passages describes manual fallback paths. The change-set clarifies that manual fallback must pass through the same governance as automated paths.	Manual fallback is permitted only through the same authentication, authorisation, attribution, timestamps, evidence, hashing, Loom logging, and applicable Governor gates as the automated paths. A manual path is never an ungoverned path; it is a path whose execution is human but whose governance, evidence, and audit controls are identical to the automated paths.	The manual fallback equivalence is now governing language. All manual operations in v14.0 pass through the Governor pipeline; the Governor records the human actor and the authorisation chain.

Appendices | Appendices A through F

#	Title	Conflict	Resolution	Integration Impact
A-19	Global Readiness Claims	The Main Blueprint includes broad "production-ready" language. The change-set establishes a deployment-state vocabulary (CORE_READY through PRODUCTION_CONNECTED and legal-authorization states) that is materially more precise.	Use the updated deployment-state vocabulary to distinguish internal architectural readiness from external operational readiness. Do not infer a working production connector merely because its adapter exists. A capability is production-operational only when it is CORE_READY, PRODUCTION_CONNECTED, and legally authorised for the relevant jurisdiction; the existence of an adapter is necessary but not sufficient. The four-dimension external readiness model (TECHNICAL, LEGAL, OPERATIONAL, COMMERCIAL) is formalised (L1, Part 6).	The deployment-state vocabulary is adopted throughout v14.0. The over-claim elimination table (Section 3) lists the forbidden phrases and their canonical replacements. The four-dimension external readiness is in L1, Part 6.
A-20	Command vs Event Semantics	The Main Blueprint in some passages conflates commands and events. The change-set establishes the formal separation of commands (requests for state change) from events (records of state change).	Commands and events are formally separated (C-22). Commands are mutable; events are immutable. Commands are issued by authenticated parties; events are recorded by the Governor with full provenance. Commands may be replayed; events are deterministic on replay. Commands are not hash-chained; events are hash-chained. The separation is enforced throughout the architecture (L1, Part 2; L1, Part 3) and the API (L2, Part 13).	The command/event separation is enshrined at L0 (C-22). The event spine (L1, Part 2) is the canonical event store. The API (L2, Part 13) enforces the separation in its command and event surfaces.

Appendices | Appendices A through F

#	Title	Conflict	Resolution	Integration Impact
A-21	Recovery vs Rollback	The Main Blueprint in some passages uses "rollback" language. The change-set clarifies that SGTX does not roll back (erase history); it recovers (builds on history).	Recovery is not rollback (C-19). The event spine is append-only; no event is ever erased. A failed state is preserved in the spine; recovery is a new sequence of events that builds on the failed state, not a replacement for it. The Recovery Vault (L2, Part 12) stores recovery materials in a content-addressable store; the failed path remains accessible for audit. Settlement reversals, where permitted, are recorded as new events (reversal events), not as erasures of the original settlement events.	The recovery-not-rollback principle is enshrined at L0 (C-19). The Recovery Vault (L2, Part 12) is the architectural realisation. The "rollback" language is eliminated in v14.0.
A-22	Proof-of-Reserves Wallet Examples	The Main Blueprint includes specific wallet address examples in the proof-of-reserves section. The change-set clarifies that such examples should be illustrative, not prescriptive, and should not imply specific custodial arrangements.	Wallet address examples in the proof-of-reserves section are illustrative, not prescriptive. The actual reserve wallets are held at licensed institutions; their addresses are published in the ZK attestation (Add-On 7) where cryptographically necessary. The examples in the documentation are placeholders; they do not represent actual reserve wallets.	The proof-of-reserves documentation (L2, Appendix A) uses illustrative examples. The actual wallet addresses are governed by the reserve policy and published via ZK attestation only where cryptographically necessary.
A-23	Legal Specificity of Jurisdiction Examples	The Main Blueprint uses Egypt as a primary jurisdiction example. The change-set clarifies that Egypt is the initial reference jurisdiction, not the only jurisdiction, and that the regulatory classifications are jurisdiction-specific.	Egypt is the initial reference jurisdiction (L1, Part 7; L2, Appendix B). The four direct Egypt integrations (Nafeza, CargoX, ETA, CBE) are the first concrete realisations of the worldwide adapter fabric. The regulatory classifications are jurisdiction-specific; the Egypt classifications (per CBE Law 194/2020 and related regulations) are the initial reference, extensible to other jurisdictions as SGTX expands.	The Egypt regulatory classification matrix (L2, Appendix B) is preserved as the initial reference. The Regulatory Classification Gate (L1, Part 7) is the architectural component that maps functionality to licence class per jurisdiction. The Egypt-as-initial-reference framing is now governing language.

Appendices | Appendices A through F

#	Title	Conflict	Resolution	Integration Impact
A-24	Incomplete Appendage of the Change-Set (Add-Ons 9-28 and Final Summary Preamble) [NEW v13.0 FINDING]	The v12.0 appended amendment body omitted Add-Ons 9 through 28 and the Final Summary preamble. The v13.0 re-audit identified this as a completeness gap.	Re-integrate the previously missing Add-Ons 9-28 and the Final Summary preamble into their canonical positions. The complete twenty-eight-add-on catalogue is now present in full (L1, Part 5). The Final Summary preamble (SGTX Master Global Trade Completion Amendment) is preserved in the source documents.	L1, Part 5 contains the complete twenty-eight-add-on catalogue with status matrix. The historical absence from v12.0 is recorded in this finding; the re-integration in v13.0 is the canonical correction. v14.0 preserves the re-integrated catalogue unchanged.

E.2 — v12.0 Audit Header (Preserved)

[L2] The original v12.0 audit header is preserved unchanged below for full traceability. Nothing from the prior audit has been removed. The v12.0 audit was carried in reverse order (A-23 down to A-01); the v13.0 re-audit re-ordered the findings into ascending logical sequence (A-01 through A-24) and added finding A-24 as a new v13.0 finding.

```
v12.0 AUDIT HEADER (PRESERVED)
=====
Audit Purpose: Re-audit of the integration of the Updates /
Modifications / Edits change-set into the SGTX Platform Main
Blueprint. The audit was carried in reverse order (A-23 down to
A-01) in v12.0 and has been re-ordered into ascending logical
sequence (A-01 through A-24) in v13.0, with finding A-24 added
as a new v13.0 finding.

Source 1 (Main Blueprint, base of integration):
SGTX_PLATFORM_MASTER_BLUEPRINT_INTEGRATED_v12.0.docx
Logical paragraph lines extracted: 76,122
Logical-text SHA-256:
c263f527f3966dab01d3cffc87e7d2747d01c017ca7a786d1964f09018087d42

Source 2 (Updates / Modifications / Edits):
sgtx add ons and modifications.rtf
Change-set text lines extracted: 7,582
Raw-RTF SHA-256:
87181d220df82a485485eec4b9896031910c79afa6332516ef2afac4682c5b72

The audit identifies twenty-four findings (A-01 through A-24).
Findings A-01 through A-23 resolve semantic and terminological
tensions between the Main Blueprint and the change-set without
deleting any material from either source. Finding A-24 corrects
a completeness gap in the v12.0 appended amendment body by
re-integrating the previously missing Add-Ons 9-28 and Final
Summary preamble.
```

E.3 — Change Log v13.1 to v14.0

[L2] The change log from v13.1 FINAL to v14.0 Clean Master Edition is preserved below. The change log is the canonical record of what changed in this edition. It is also captured in Section 8 of this document.

Appendices | Appendices A through F

#	Change	Rationale
1	Three-layer separation (L0/L1/L2).	Distinguishes immutable constitutional principles from normative architecture and implementation specifications.
2	Over-claim elimination: "production-ready" becomes CORE_READY / PRODUCTION_CONNECTED.	Resolves finding A-19 textually; distinguishes internal architectural readiness from external operational readiness.
3	28-add-on status matrix added.	Provides per-add-on status visibility per the deployment-state vocabulary.
4	Four-dimension external readiness added (TECHNICAL, LEGAL, OPERATIONAL, COMMERCIAL).	Formalises the jurisdiction readiness model; no dimension may be inferred from another.
5	Command/Event taxonomy formalised.	Elevates finding A-20 to a first-class architectural principle (C-22).
6	All v13.1 Part A resolutions made governing language.	The v13.0 audit resolutions are no longer advisory; they are governing in v14.0.
7	"complete" replaced with "specified".	Avoids implying implementation completeness where only specification exists.
8	"zero-cost" replaced with "institutional-cost scope clarified".	Acknowledges that all integrations carry institutional cost (finding A-15).
9	"globally available" replaced with "jurisdiction-aware, capability-gated".	Reflects that availability is per-jurisdiction and per-capability.
10	Recovery-not-rollback principle enshrined (C-19).	Eliminates "rollback" language; recovery builds on history (finding A-21).
11	AI Authority Ladder clarified: A4 is deterministic policy automation, not AI autonomy.	Resolves finding A-02 textually.
12	Non-custody is architectural, not legal classification.	Resolves finding A-03 textually.
13	USTN as namespace, not override of external authority.	Resolves finding A-12 textually.
14	Egypt as initial reference jurisdiction, not the only jurisdiction.	Resolves finding A-23 textually.
15	RoRo as first-class transport mode.	Resolves finding A-09 textually.
16	Audit trail (Appendix E) preserved verbatim.	Honours the immutability of the audit trail; nothing from v13.1 is removed.

Appendix F — Source Manifest

[L2] The Source Manifest identifies the source documents integrated into this Clean Master Edition. Two primary sources were integrated: the v12.0 Main Blueprint and the Add-Ons Change-Set. The SHA-256 hashes below are the cryptographic fingerprints of the source documents; they enable verification that the sources used to produce this edition are the sources referenced.

Source 1 — Main Blueprint (base of integration)

Field	Value
Filename	SGTX_PLATFORM_MASTER_BLUEPRINT_INTEGRATED_v12.0.docx
Description	The SGTX Platform Master Blueprint, v12.0 integrated edition. This source contains an earlier (v12.0) integration pass and is treated as the authoritative historical/architectural body of the blueprint.
Logical paragraph lines extracted	76,122
Logical-text SHA-256	c263f527f3966dab01d3cffc87e7d2747d01c017ca7a786d1964f09018087d42
Role in v14.0	Preserved as the historical body; superseded by v14.0's three-layer structure where normative. Source material is referenced in Appendix E and throughout L1 and L2.

Source 2 — Updates / Modifications / Edits (Change-Set)

Field	Value
Filename	sgtx add ons and modifications.rtf
Description	The SGTX Platform Complete Add-Ons Specification covering Add-Ons 1 through 28, followed by a Final Summary that includes the SGTX Master Global Trade Completion Amendment.
Change-set text lines extracted	7,582
Raw-RTF SHA-256	87181d220df82a485485eec4b9896031910c79afa6332516ef2afac4682c5b72
Role in v14.0	Preserved as the add-on catalogue and amendment body; integrated into L1, Part 5 (Add-On Catalogue) and reflected throughout L0 and L1 as governing language.

Source Hash Verification

[L2] To verify that this edition was produced from the canonical sources, compute the SHA-256 hashes of the source files and compare them to the hashes above. The hashes are computed over the raw bytes of the source files (raw-RTF for Source 2; logical-text for Source 1 as extracted by the integration pipeline). Mismatches indicate that the source has been modified after integration; such mismatches require re-integration before the edition can be considered authoritative.

```
# SHA-256 verification commands (Linux/macOS)
sha256sum sgtx_add_ons_and_modifications.rtf
# Expected: 87181d220df82a485485eec4b9896031910c79afa6332516ef2afac4682c5b72

# For Source 1 (logical text extraction), use the integration pipeline:
# python3 integration_pipeline.py extract-text \
# SGTX_PLATFORM_MASTER_BLUEPRINT_INTEGRATED_v12.0.docx | sha256sum
# Expected: c263f527f3966dab01d3cffc87e7d2747d01c017ca7a786d1964f09018087d42
```

Source Provenance

[L2] Both sources were provided to the Master Blueprint Integration Engine as authoritative inputs. The integration process is recorded in the Loom audit log; the integration artefacts (extracted text, normalised text, integrated master) are themselves content-addressable and hash-verified. The integration is reproducible: re-running the integration pipeline on the same sources produces the same integrated master (modulo editorial changes introduced by the integration engine, which are themselves logged).

SECTION 8

Change Log v13.1 to v14.0

The canonical record of what changed in this edition.

This section is the canonical record of changes from v13.1 FINAL to v14.0 Clean Master Edition. The change log is preserved as a governing instrument: the changes recorded here are the authoritative description of the v14.0 edition's delta from v13.1. The change log is also preserved in Appendix E.3 for traceability.

8.1 — Three-Layer Separation (L0/L1/L2)

The defining change in v14.0 is the introduction of an explicit three-layer separation. Every normative statement in the document is now tagged with its layer: [L0] for immutable constitutional principles, [L1] for normative architecture, [L2] for implementation specifications. This separation enables auditors, regulators, and engineers to determine at a glance which statements are immutable (and thus require constitutional amendment to change) versus which are architectural or implementation-level (and thus may evolve through governed change-management).

8.2 — Over-Claim Elimination

The over-claim vocabulary carried over from the v12.0 Main Blueprint is eliminated. The forbidden phrases and their canonical replacements are listed in Section 3 (Executive Summary, Over-Claim Elimination). The most significant replacement is "production-ready" becomes "CORE_READY" or "PRODUCTION_CONNECTED", distinguishing internal architectural readiness from external operational readiness. This resolves finding A-19 textually: the deployment-state vocabulary is now used consistently throughout the document.

8.3 — 28-Add-On Status Matrix Added

The 28-add-on catalogue (L1, Part 5) now includes a status matrix showing each add-on's deployment state (CORE_READY, PRODUCTION_CONNECTED, LEGAL_AUTHORIZATION_REQUIRED, or FOUNDATION), priority (P2, P3, or P4 per Appendix D), blueprint reference, and non-marketplace bound (NMB) indicator. The matrix provides at-a-glance visibility into the deployment state of each add-on, replacing the unstructured prose descriptions in v13.1.

8.4 — Four-Dimension External Readiness Added

The four-dimension external readiness model (TECHNICAL, LEGAL, OPERATIONAL, COMMERCIAL) is formalised in L1, Part 6 (Jurisdiction Capability Adapter Schema). The model evaluates each jurisdiction's external readiness along four independent dimensions; no dimension may be inferred from another. This operationalises the deployment-state vocabulary (CORE_READY / PRODUCTION_CONNECTED / LEGAL_AUTHORIZATION_REQUIRED) into a concrete evaluation framework.

Section 8 | Change Log v13.1 to v14.0

8.5 — Command/Event Taxonomy Formalised

The command/event separation (finding A-20) is elevated from an audit resolution to a first-class architectural principle, enshrined as C-22 in the L0 constitution. The L1 architecture (Part 2, Event Spine; Part 3, Governor Pipeline) and the L2 implementation (Part 11, Event Type Catalogue; Part 13, API Contract) all enforce the separation. Commands are mutable; events are immutable. Commands request state change; events record state change.

8.6 — All v13.1 Part A Resolutions Made Governing Language

The twenty-four findings of the v13.0 audit (A-01 through A-24) were advisory resolutions in v13.1. In v14.0, they are governing language: the resolutions are encoded as L0 constitutional statements, L1 architectural commitments, or L2 implementation specifications, as appropriate. The audit trail (Appendix E) preserves the original findings and resolutions verbatim; the governing-language versions are in the main body.

8.7 — Additional Replacements

Forbidden Phrase	Canonical Replacement	Finding Resolved
complete	specified	General over-claim elimination.
zero-cost	institutional-cost scope clarified	A-15.
globally available	jurisdiction-aware, capability-gated	A-19, A-23.
rollback	recovery	A-21.
marketplace (colloquial)	trade execution infrastructure	A-16.
stablecoin (SGTX-issued)	USTN-backed reserve instrument	A-04.
DeFi integration	(forbidden; no replacement)	A-04.
A4 Governance / auto-executes	A4: Execution Within Bounds (deterministic policy automation)	A-02.
multi-region mirroring (for EGYPT_ONLY)	in-region replication	A-14.
Direct API (as a limit)	Direct API (as a current adoption)	A-01.
Seven critical add-ons (as the catalogue)	Twenty-eight-add-on catalogue (seven as historical reference)	A-08.
RoRo as Ocean submode	RoRo as first-class transport mode	A-09.
Nafeza as one generic integration	Nafeza as one connector with mode-specific adapters	A-10.
ISO 20022 as canonical data	ISO 20022 as integration output	A-13.
production-ready	CORE_READY or PRODUCTION_CONNECTED	A-19.

SECTION 9

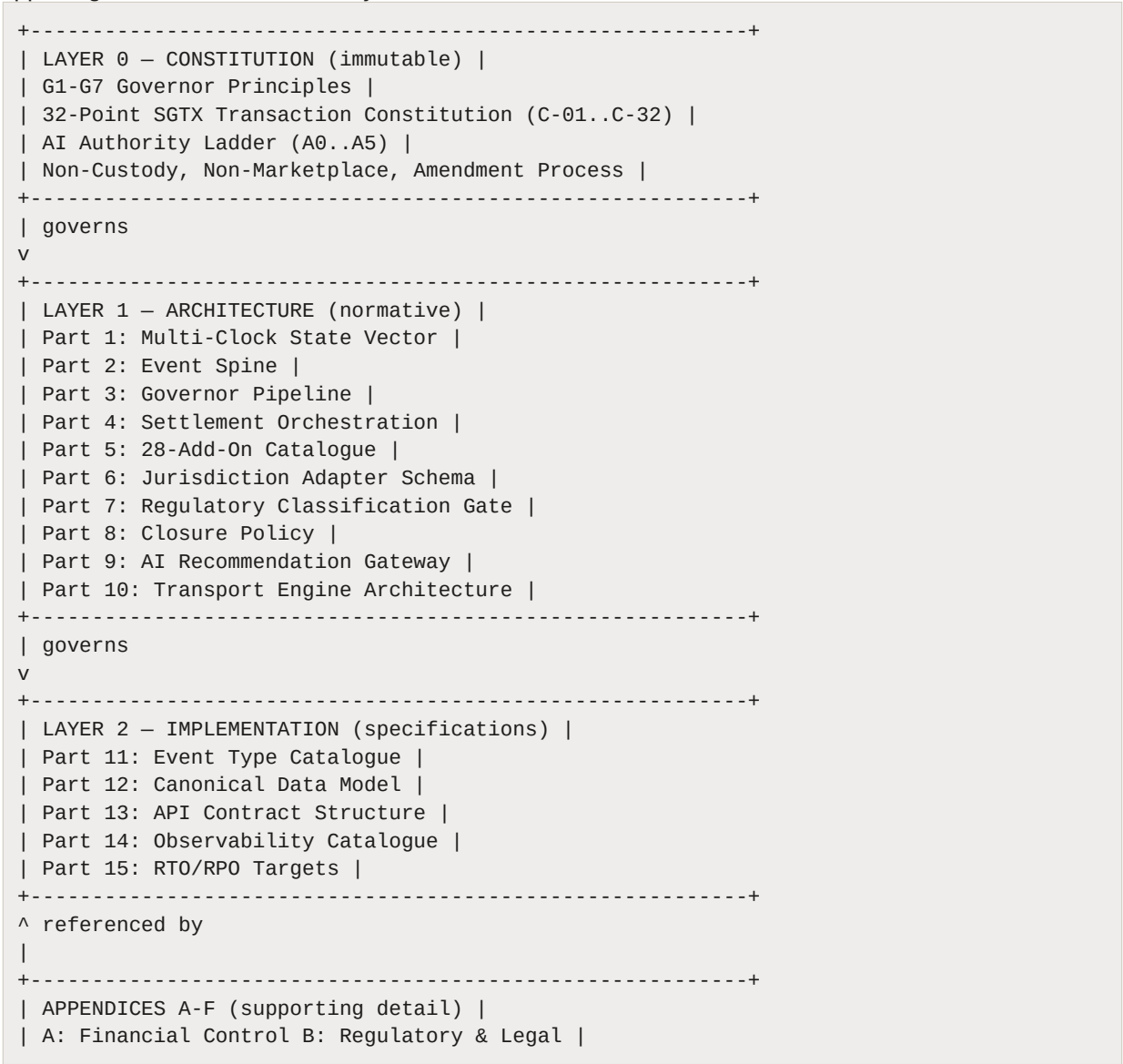
Dependency Graph of Major Components

Visual diagram showing the structural dependencies between the constitution, architecture, implementation, and appendices, and the major component groups.

The dependency graph below shows the structural dependencies between the major components of the SGTX platform. The graph is organised into three layers (Constitution, Architecture, Implementation) plus the Appendices, with three major component groups (Governor stack, Event Spine stack, Settlement stack) shown as sub-graphs.

9.1 — Layer Dependency Graph

The three layers form a strict dependency hierarchy: Constitution governs Architecture; Architecture governs Implementation. No layer may override or weaken a higher layer. The Appendices provide supporting detail across all three layers.



Section 9.1 Dependency Graph of Major Components

```

| C: Operating Model D: Implementation Priority |
| E: Audit Trail F: Source Manifest |
+-----+

```

9.2 — Governor Stack

The Governor stack enforces the constitution (L0) through the pipeline: Governor receives commands, OPA evaluates policy, WasmEdge vetoes unconstitutional decisions, Loom records every decision. The stack is the operational enforcement mechanism for G1 through G7.

```

+-----+
| Command (incoming) |
+-----+-----+
|
v
+-----+-----+
| GOVERNOR (G1) | Execution Always Gated
+-----+-----+
|
v
+-----+-----+
| OPA Policy (G2) | Policy as data
+-----+-----+
|
v
+-----+-----+
| WasmEdge Const. (G3) | Constitutional hard boundary
+-----+-----+
|
+-----+-----+
| | |
v v v
+-----+ +-----+ +-----+ +-----+
| Multisig | | AI Adv. | | Bank Auth. |
| Check (G5) | | Input (G6) | | Settle (G7)|
+-----+ +-----+ +-----+ +-----+
| | |
+-----+-----+
|
v
+-----+-----+
| LOOM AUDIT (G4) | Append-only, hash-chained
+-----+

```

9.3 — Event Spine Stack

The Event Spine stack is the authoritative history. Commands flow into the Governor; the Governor's decisions and resulting state changes are recorded as events on the spine. The State Vector Store is derived from the spine; the Transaction Twin is derived from the spine and the State Vector Store; the Closure Policy evaluates against the State Vector Store.

```

+-----+ +-----+
| Commands | -----> | Governor |
+-----+ +-----+
|
v
+-----+ +-----+
| Recovery Vault |<---| Event Spine | (immutable,
| (content-addr.) | | (hash-chained)| hash-chained,

```

Section 9.1 Dependency Graph of Major Components

```

+-----+ +-----+ +-----+ quorum-replicated)
|
v
+-----+
| State Vector | (derived view:
| Store | 12 domains x F0-F5)
+-----+
|
v
+-----+
| Transaction | (digital twin:
| Twin (14 dom.) | queryable)
+-----+
|
v
+-----+
| Closure Policy | canClose()
| (Part 8) | predicate
+-----+

```

9.4 — Settlement Stack

The Settlement stack orchestrates multi-leg settlement. The Bank Settlement Gateway interfaces with executing banks; the Settlement Orchestration Control Plane coordinates the legs; Payment Legs are the individual payment obligations. The non-custody boundary is enforced: SGTX orchestrates; banks execute; SGTX never holds funds.

```

+-----+ +-----+
| Settlement | | Payment Leg |
| Instruction |--->| (payer, payee, |
| (atomicity policy) | | amount, currency) |
+-----+ +-----+
| |
v v
+-----+ +-----+
| Settlement Orchestration Control Plane |
| (Part 4) |
+-----+ +-----+
| |
v v
+-----+ +-----+
| Bank Settlement | | Settlement |
| Gateway | | Confirmation |
| (6-stage pipeline: | | (bank-issued, |
| schema -> sig -> |--->| recorded as |
| USTN -> bene -> | | Confirmation event) |
| policy -> AML) | | |
+-----+ +-----+
|
| non-custody boundary
v
+-----+ +-----+
| Executing Bank |--->| Fiat Movement |
| (licensed, per leg) | | (bank-authoritative) |
+-----+ +-----+
^
| SGTX orchestrates instructions;
| SGTX NEVER holds funds (C-01, C-21, G7).

```

9.5 — Summary

The dependency graph illustrates the constitutional discipline of the SGTx platform: every component has a defined place in the layer hierarchy; every dependency flows in the direction of constitutional authority (L0 governs L1 governs L2); the three major component groups (Governor stack, Event Spine stack, Settlement stack) each enforce a distinct aspect of the constitution. The graph is not a runtime call graph; it is a normative dependency graph showing which components depend on which others for their constitutional authority.

Section 9 | Dependency Graph of Major Components

END OF DOCUMENT

SGTX Platform Master Blueprint — Clean Master Edition v14.0
Sovereign Governed Trade Execution Infrastructure
Audited / Integrated / Canonical

Prepared by Master Blueprint Integration Engine — 2026-08-26